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USING TIME SERIES ANALYSIS FOR SALES AND DEMAND FORECASTING

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1. INTRODUCTION

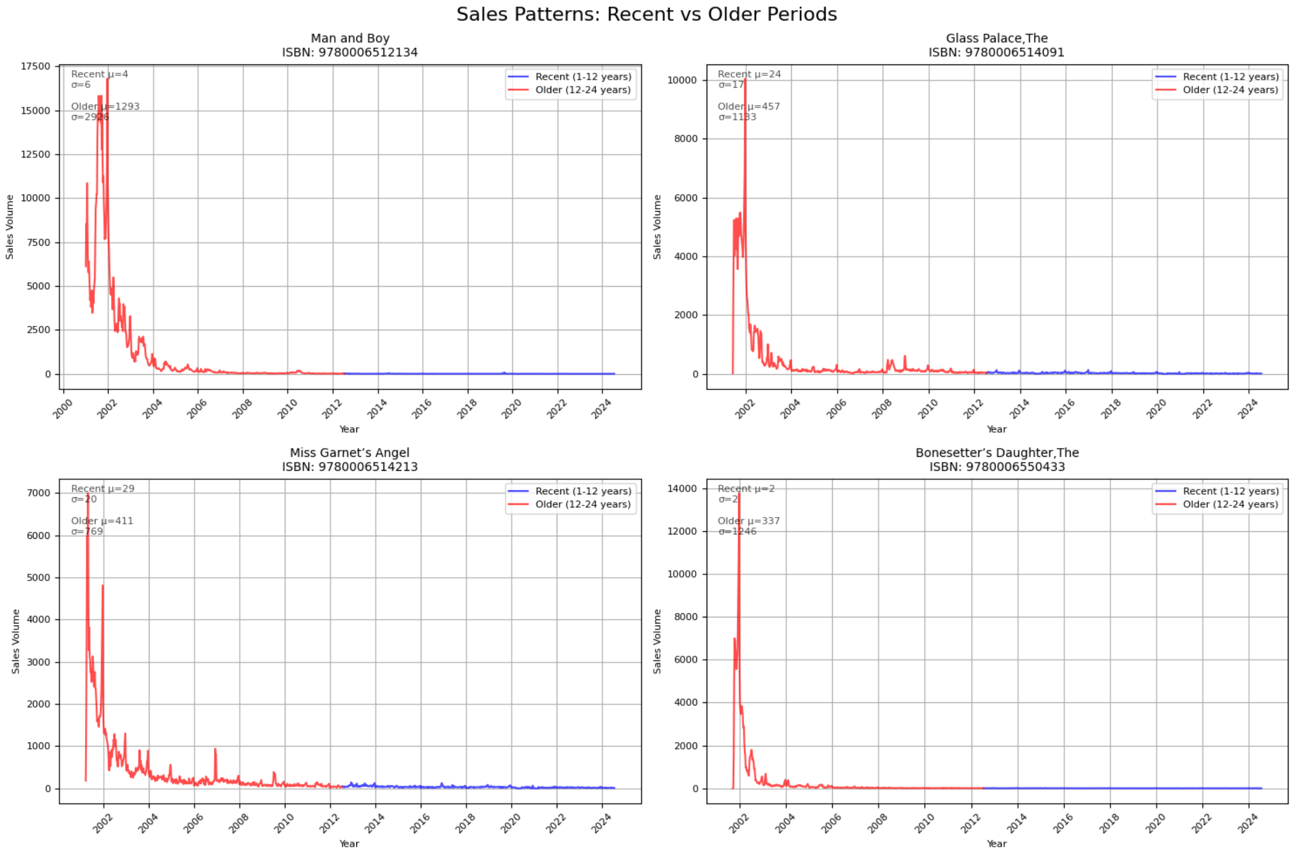
This report presents a comprehensive time series analysis of Nielsen's book sales data, focusing on identifying seasonal patterns and sales trends to support small to medium-sized publishers in making data-driven decisions.

Using datasets containing detailed sales information across four major book categories - Adult Fiction, Adult Non-Fiction (Specialist and Trade), and Children's/YA literature - we analysed over 227,000 weekly sales records for 500 unique ISBNs. Our analysis employs various forecasting models including ARIMA, XGBoost, and LSTM to predict sales patterns and identify books with strong long-term potential. The findings aim to optimise inventory management and improve publishers' ability to identify titles with sustainable market performance, ultimately helping to balance initial investment decisions with ongoing stock control.

2. DATA PREPROCESSING AND INITIAL ANALYSIS

The analysis of Nielsen's book sales data spanning 2001-2024 began with comprehensive preprocessing, including standardising book titles, converting ISBNs, and resampling weekly data with zero-fill for missing weeks. From the initial 500 ISBNs

Fig 1: General Sales Pattern



across four major categories, 61 titles were identified as having significant long-term sales patterns worthy of deeper analysis (See Fig 1 and more in Appendix A).

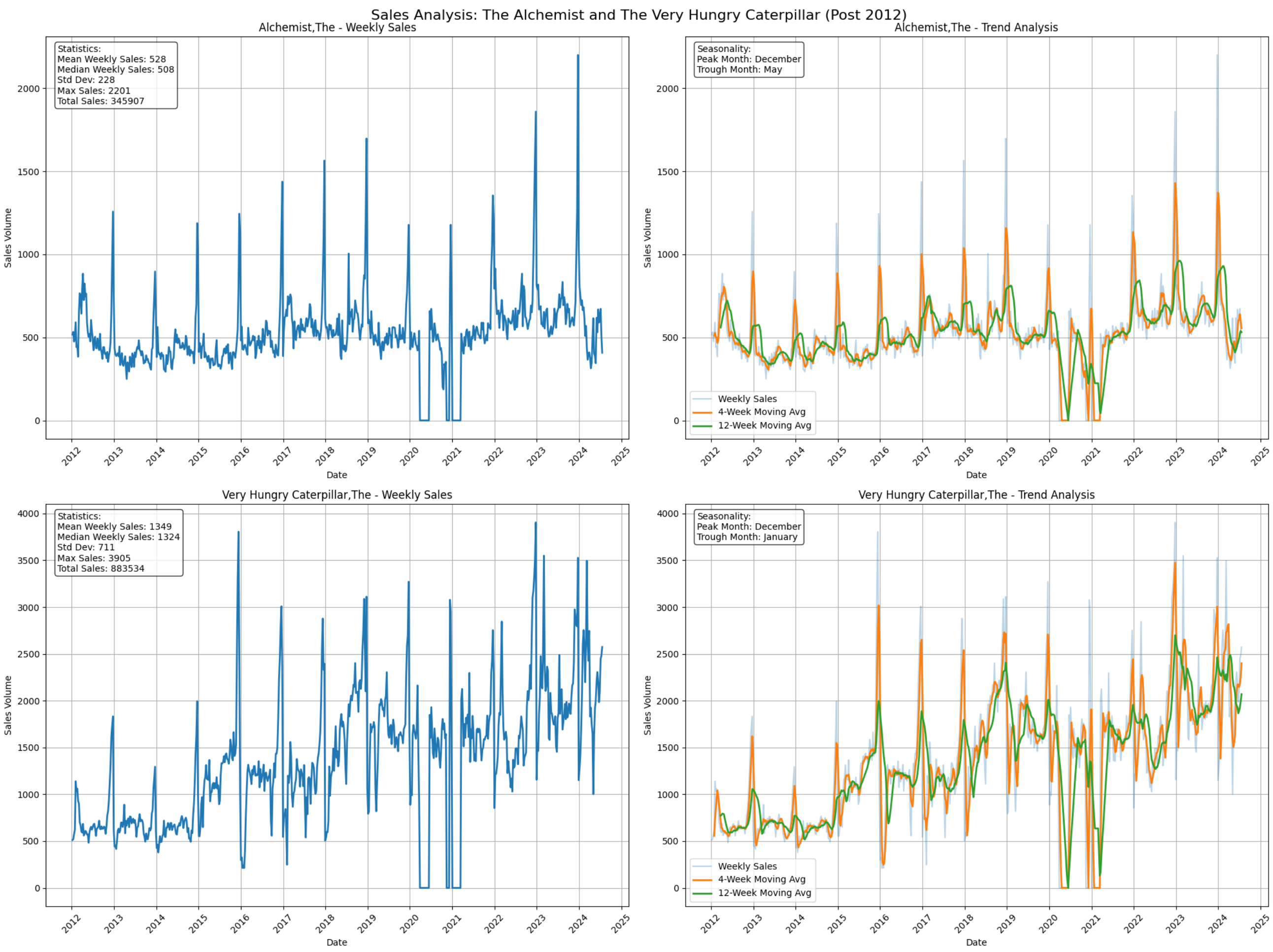
Comparative analysis of sales patterns between 1-12 and 12-24 year periods revealed a marked evolution in the book market. Earlier periods showed higher initial sales peaks and volatility, while recent years demonstrate more stable volumes with gradual decline patterns, reflecting the industry's adaptation to digital transformation and evolving consumer behaviours.

FOCUS ON THE ALCHEMIST AND THE VERY HUNGRY CATERPILLAR

"The Very Hungry Caterpillar" emerged as a standout performer with remarkable weekly sales averaging 1,349 copies and total sales of 883,534 units during 2012-2024. The title exhibited strong seasonality with December peaks and January troughs, alongside a significant upward trend starting 2015-2016. This performance notably contrasts with more typical sales decay patterns seen across the dataset.

"The Alchemist" displayed different but equally interesting characteristics, maintaining steady moderate demand with average weekly sales of 528 copies and total sales of 345,907 units. Its stability is reflected in a lower coefficient of variation (43.17% vs

Fig 2: Focus on The Alchemist and The Very Hungry Caterpillar



52.69% for Caterpillar), though both titles show consistent base demand and clear seasonal patterns, particularly during the holiday season.

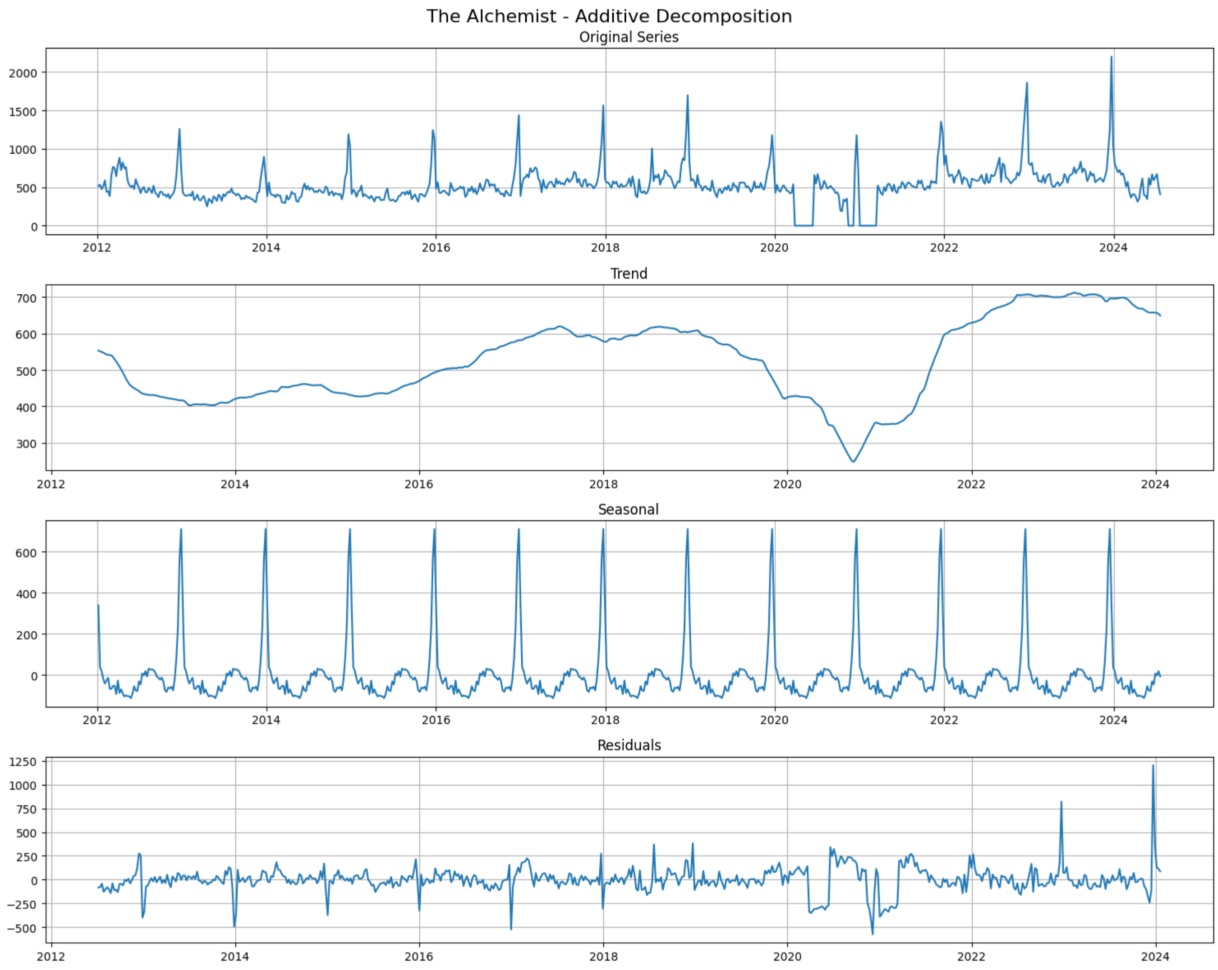
Both books demonstrated exceptional longevity, experiencing their peak sales in recent years (2022/2023), defying the typical long-term decay pattern observed in most titles. This analysis informed our selection of these titles for more detailed forecasting model development.

3. TIME SERIES ANALYSIS & FORECASTING

3.1. DEEP DIVE: THE ALCHEMIST & VERY HUNGRY CATERPILLAR

Time series decomposition revealed distinct characteristics for both titles. The Alchemist maintained a stable trend (See Fig 3) with mean sales of 524 copies (SD=112), showing resilience through market changes except for a U-shaped pattern

Fig 3: The Alchemist Time series Decomposition Analysis



during 2020-2022. Its seasonal component displayed consistent annual cycles with December peaks averaging 600 units above trend. The Very Hungry Caterpillar demonstrated (See B.1) more dynamic behaviour with higher mean sales of 1,341 copies (SD=443) and stronger upward progression, particularly post-2021.

Stationarity analysis indicated both series (See B2 and B3) required regular and seasonal differencing for optimal modelling. The Very Hungry Caterpillar showed higher volatility and heteroscedasticity, particularly in recent years, while The Alchemist exhibited more stable variance structures after differencing.

ACF/PACF analysis revealed (See B4 and B5) strong first-lag correlations for both titles (0.76 for Alchemist, 0.85 for Caterpillar) with significant seasonal components at lag 52. The Alchemist's patterns suggested an AR(1) process with clear yearly seasonality, while Caterpillar showed a more complex AR structure with multiple influential periods, indicating more sophisticated seasonal dynamics requiring careful model specification.

3.2. MODEL DEVELOPMENT & RESULTS

We implemented multiple forecasting approaches with varying levels of success. Auto ARIMA identified optimal models for both titles: SARIMAX(1,1,3)x(0,1,1)[52] for The Alchemist and SARIMAX(3,1,1)x(2,1,0)[52] for The Very Hungry Caterpillar, achieving MAPEs of 27.15% and 16.60% respectively for 32-week forecasts. See Fig 4 and Fig 5.

Fig 4: Auto ARIMA Forecast - The Alchemist

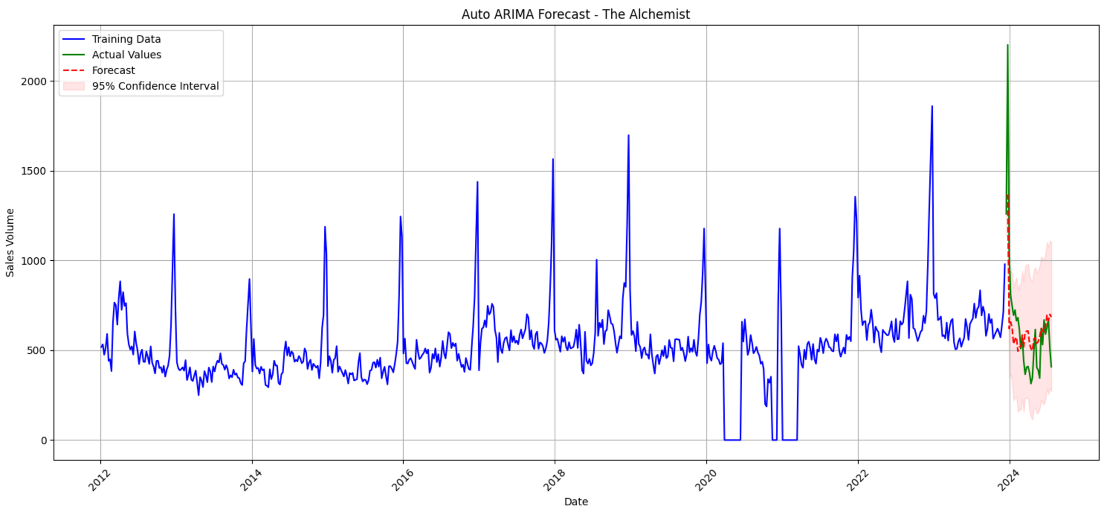
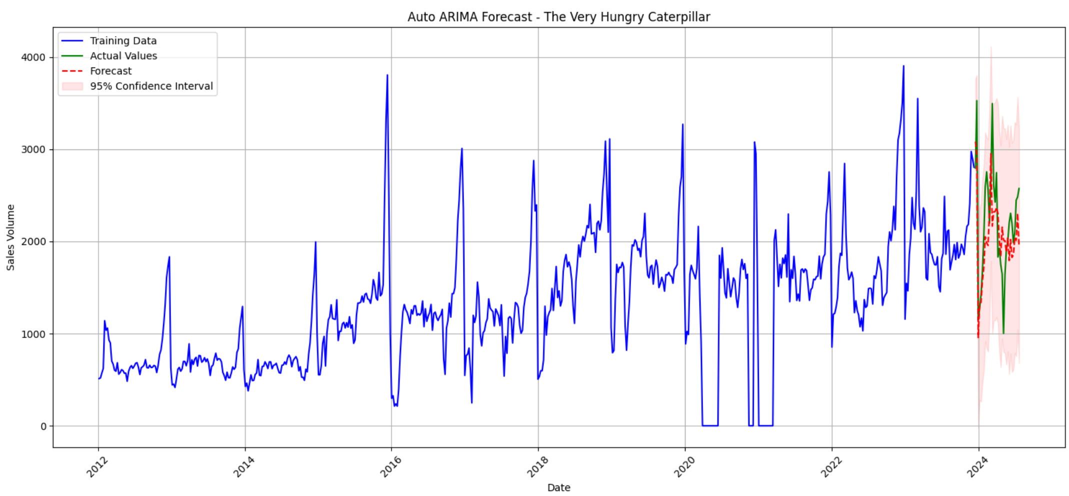
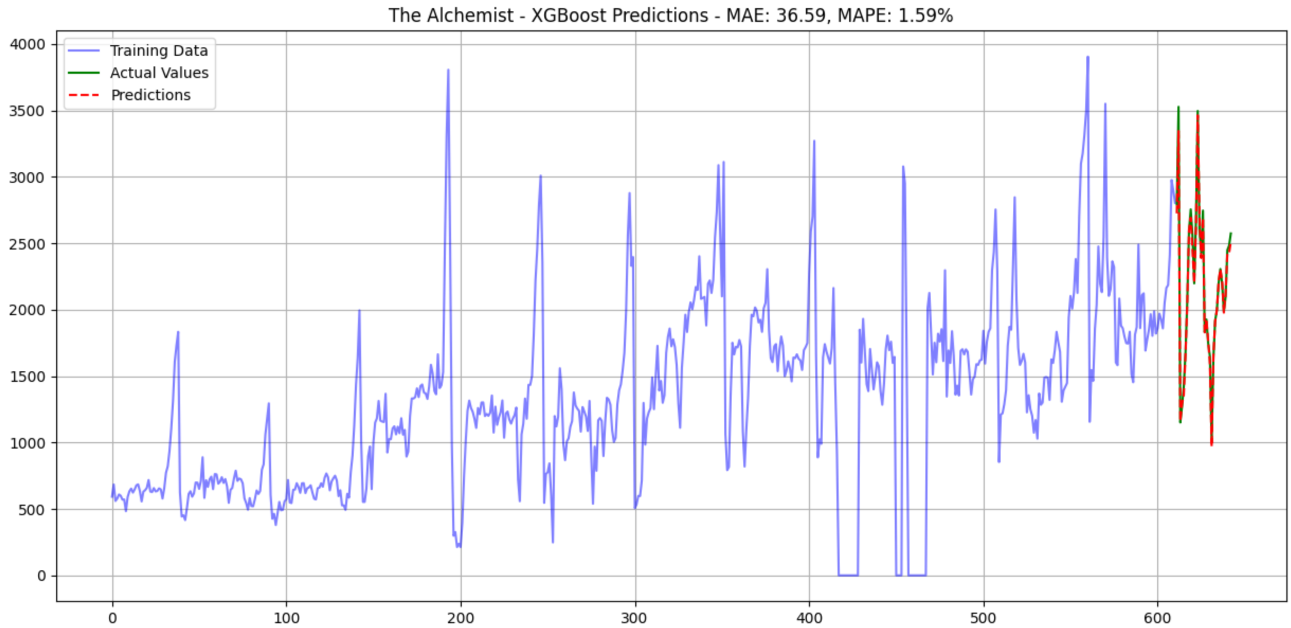


Fig 5: Auto ARIMA Forecast - The Very Hungry Caterpillar



XGBoost demonstrated exceptional predictive accuracy, achieving MAPE of 1.59% for The Alchemist and 1.89% for The Very Hungry Caterpillar (See C1). The visualisations show the model's ability to capture both long-term trends and seasonal fluctuations accurately, with predictions (red) closely tracking actual values (green). Feature importance analysis revealed that month (f0) and week (f1) were the dominant predictors, followed by one-week lag values (f6), suggesting that recent sales history and seasonal timing are the most crucial factors for accurate forecasting.



LSTM models (see C3) showed moderate performance (MAPE ~19% for both books) despite architectural tuning. Hybrid approaches (see D1) combining SARIMA and LSTM were explored, with parallel implementation (see D4) outperforming sequential. The Alchemist performed best with 30% SARIMA/70% LSTM weight, while Caterpillar favoured 70% SARIMA/30% LSTM.

Weekly forecasting consistently outperformed monthly aggregation across all models. While monthly XGBoost predictions showed MAPEs of 25.51% and 21.01% respectively (see E1 and E4), weekly predictions maintained superior accuracy, suggesting the importance of granular temporal patterns in book sales forecasting.

4. PERFORMANCE EVALUATION & CONCLUSIONS

XGBoost consistently demonstrated superior performance across both titles, with MAPE under 2% for weekly forecasts. Traditional SARIMA models proved more reliable for monthly predictions of high-volume titles like The Very Hungry Caterpillar (MAPE: 20.26%), while hybrid approaches offered balanced but not superior performance.

Analysis revealed distinct seasonal patterns requiring different inventory strategies. The Very Hungry Caterpillar shows pronounced December peaks (see B1) requiring substantial holiday stock adjustment, while The Alchemist maintains more consistent demand patterns (see Fig 3) suggesting stable base stock levels.

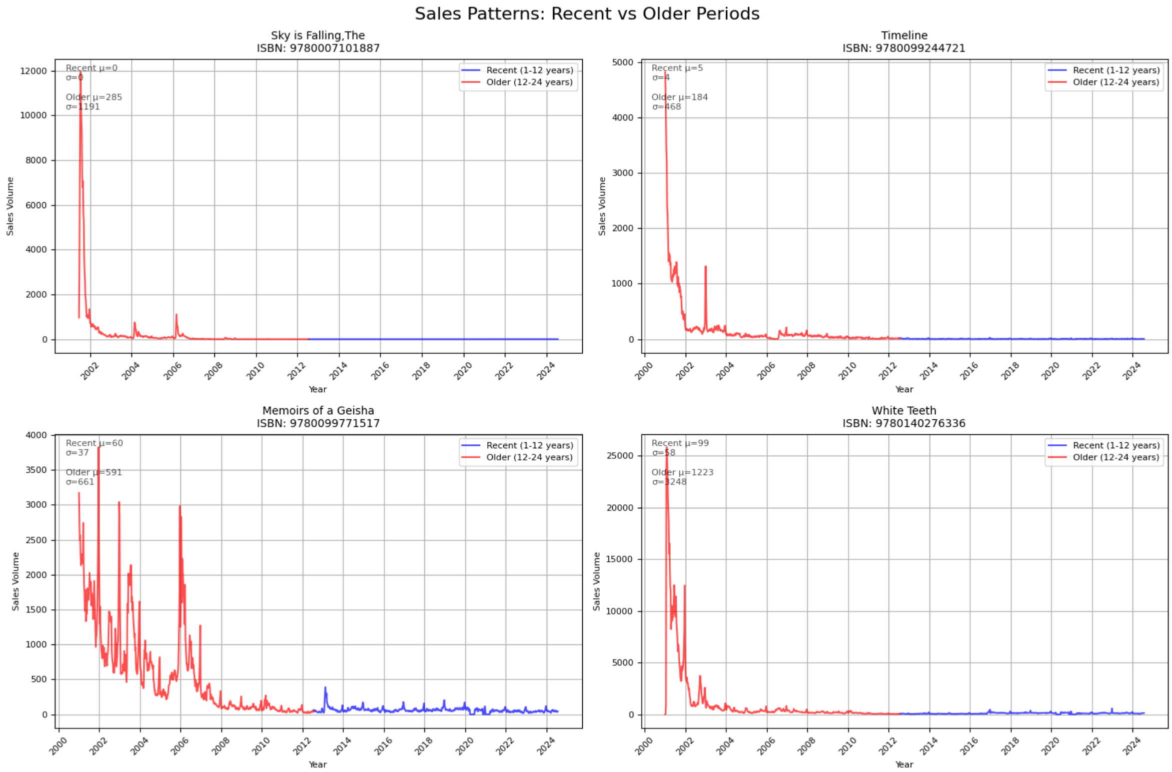
For implementation, we recommend:

- Utilising weekly XGBoost models for short-term inventory decisions
- Maintaining separate models for children's books vs adult titles given their distinct patterns
- Implementing higher safety stocks for December-January period
- Regular model retraining incorporating COVID-19 impact
- Consider parallel hybrid models for long-term strategic planning where stability is prioritised over precision

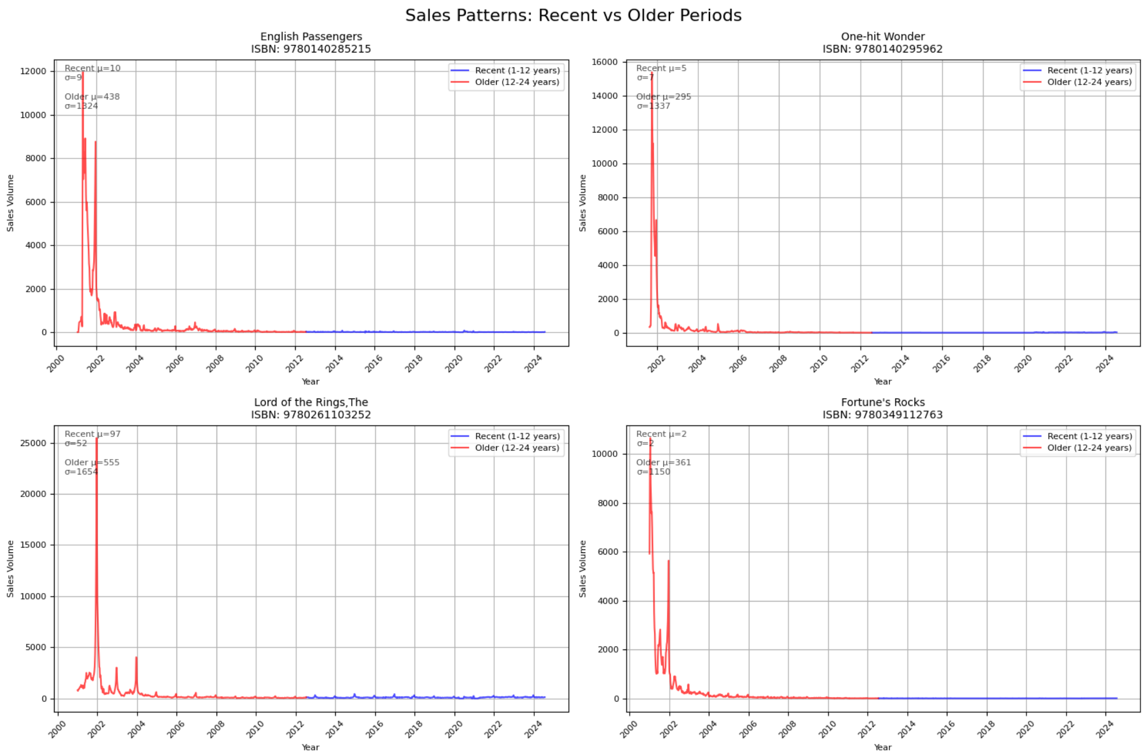
APPENDIX

A: EDA

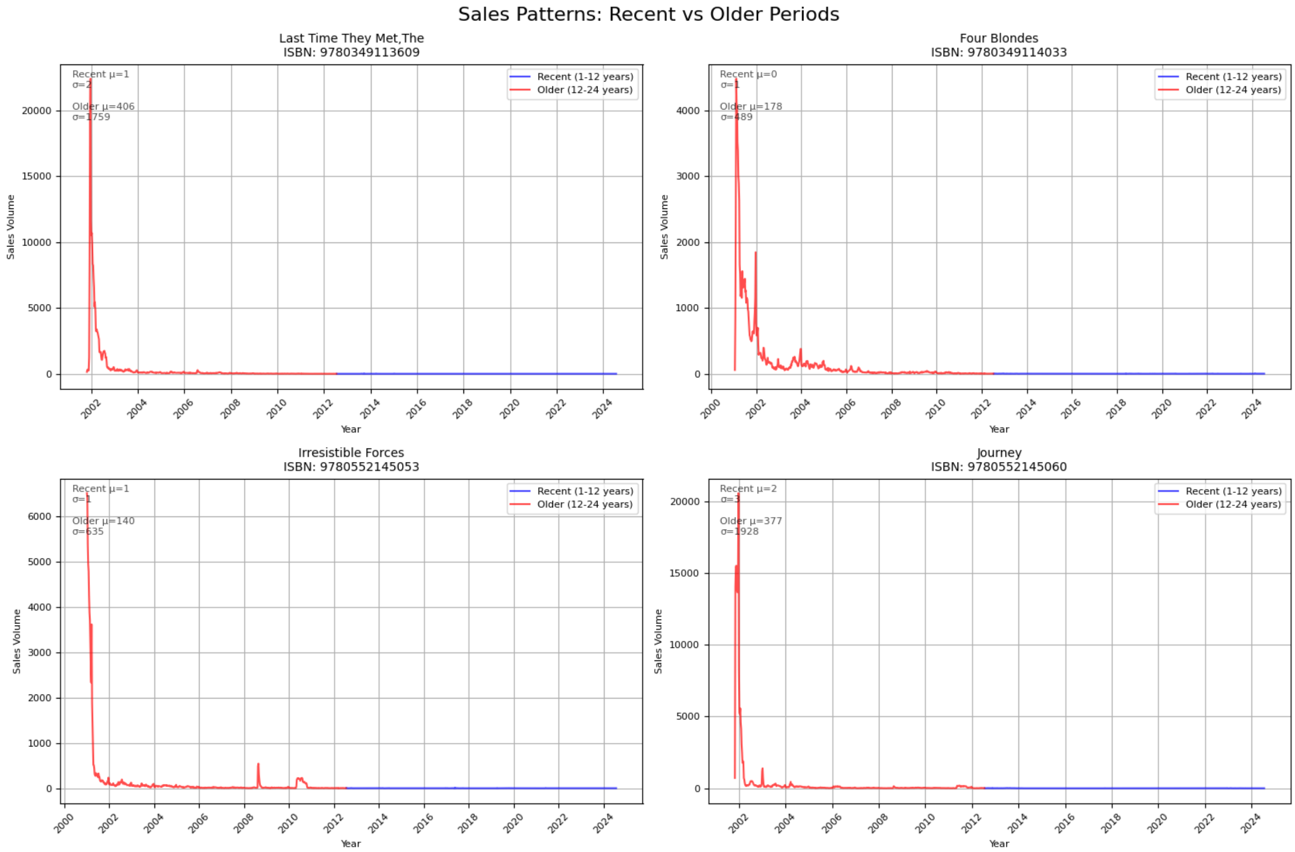
A.1: Sales Patterns



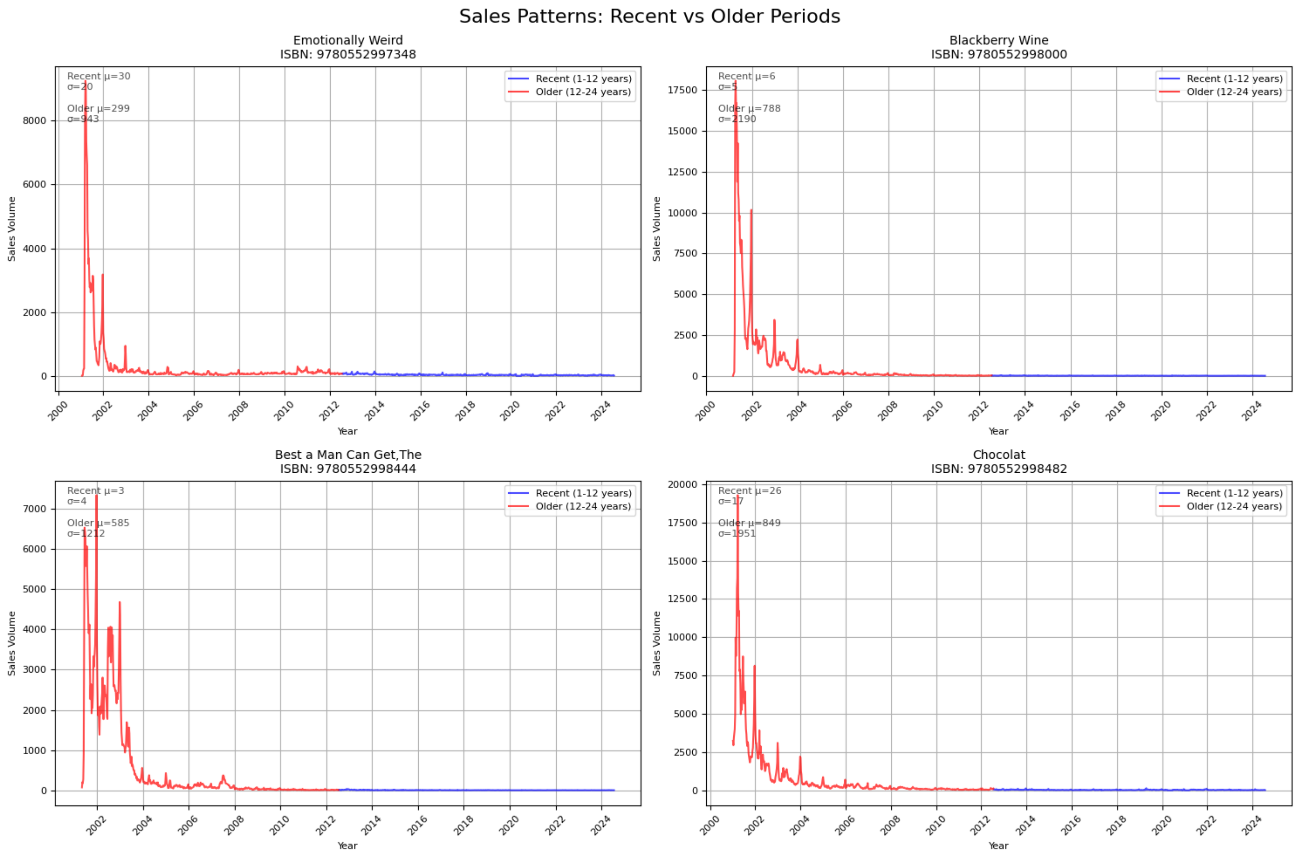
A.2: Sales Patterns



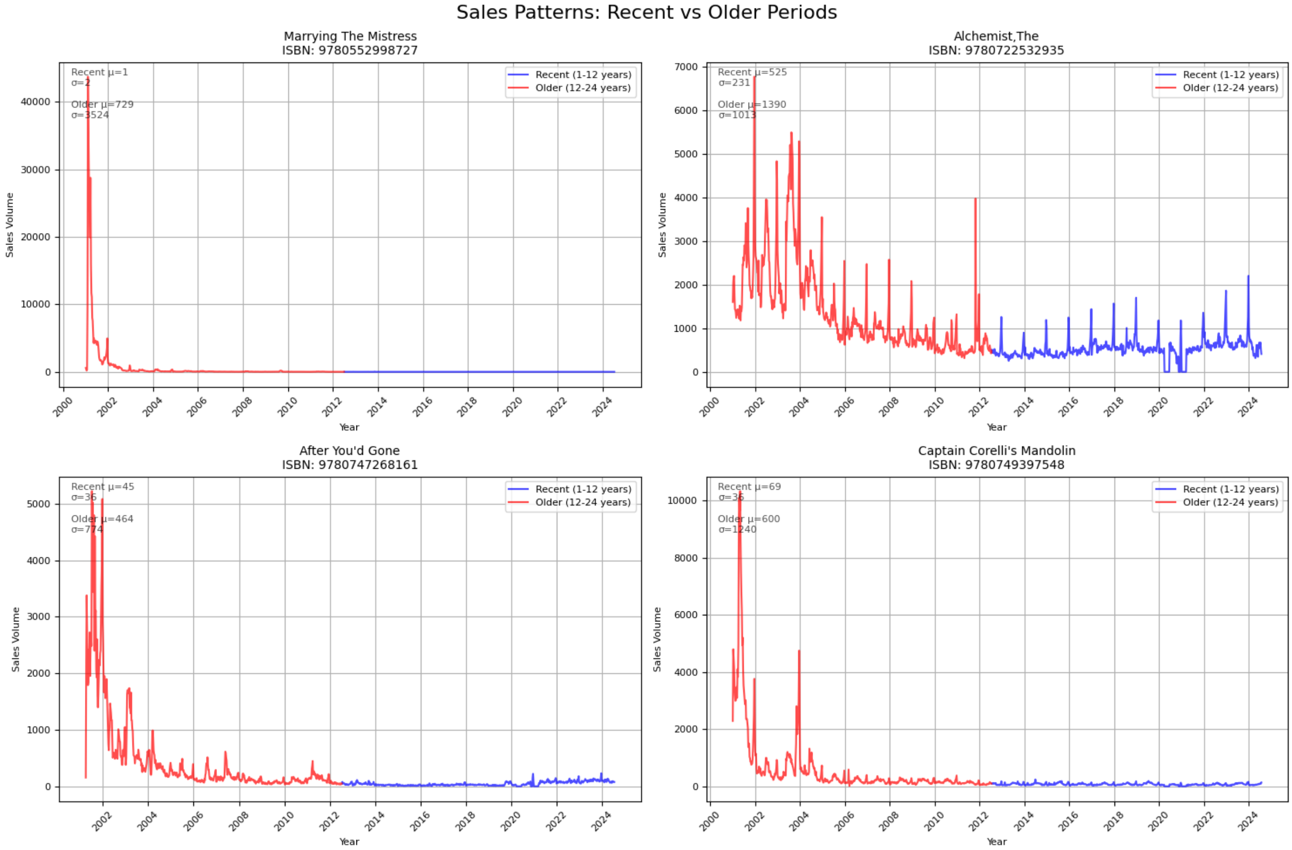
A.3: Sales Patterns



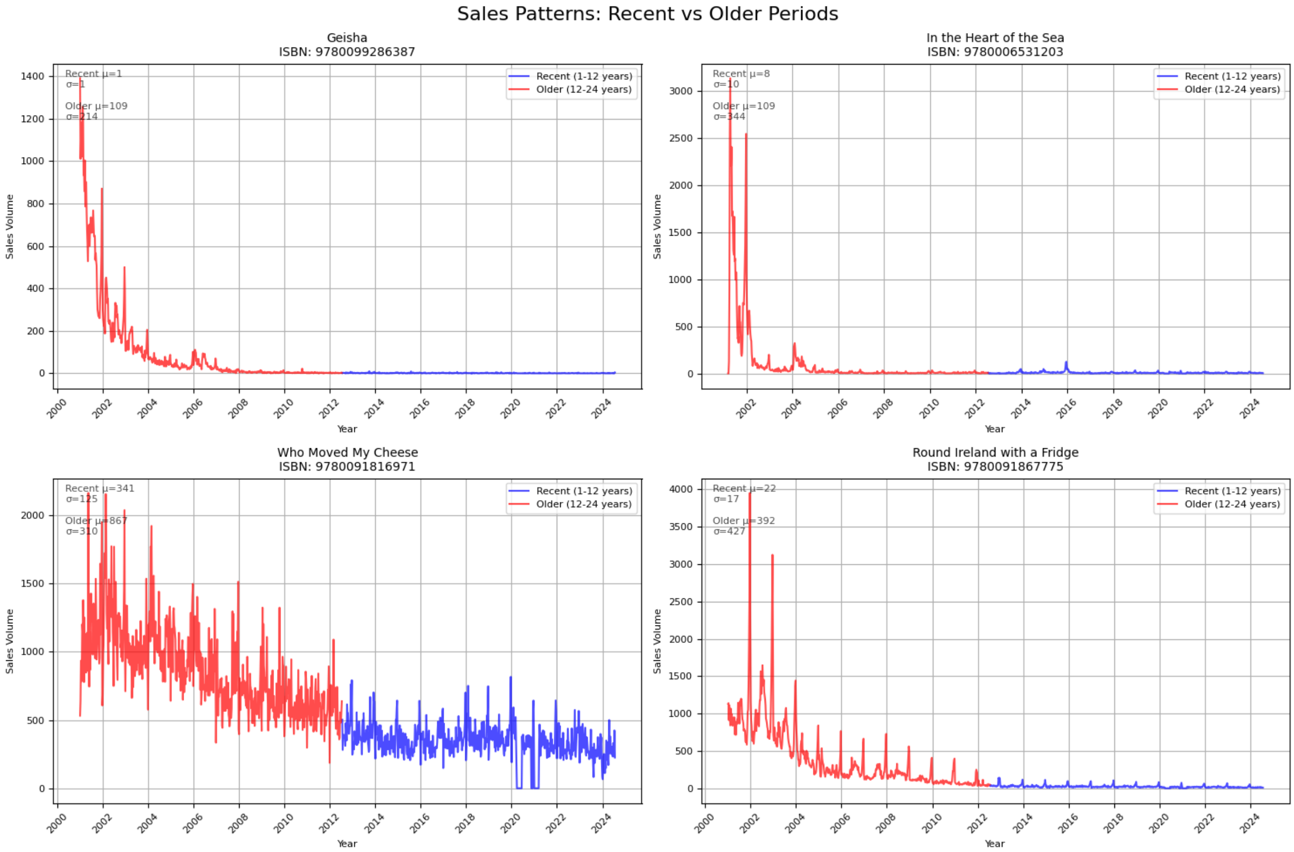
A.4: Sales Patterns



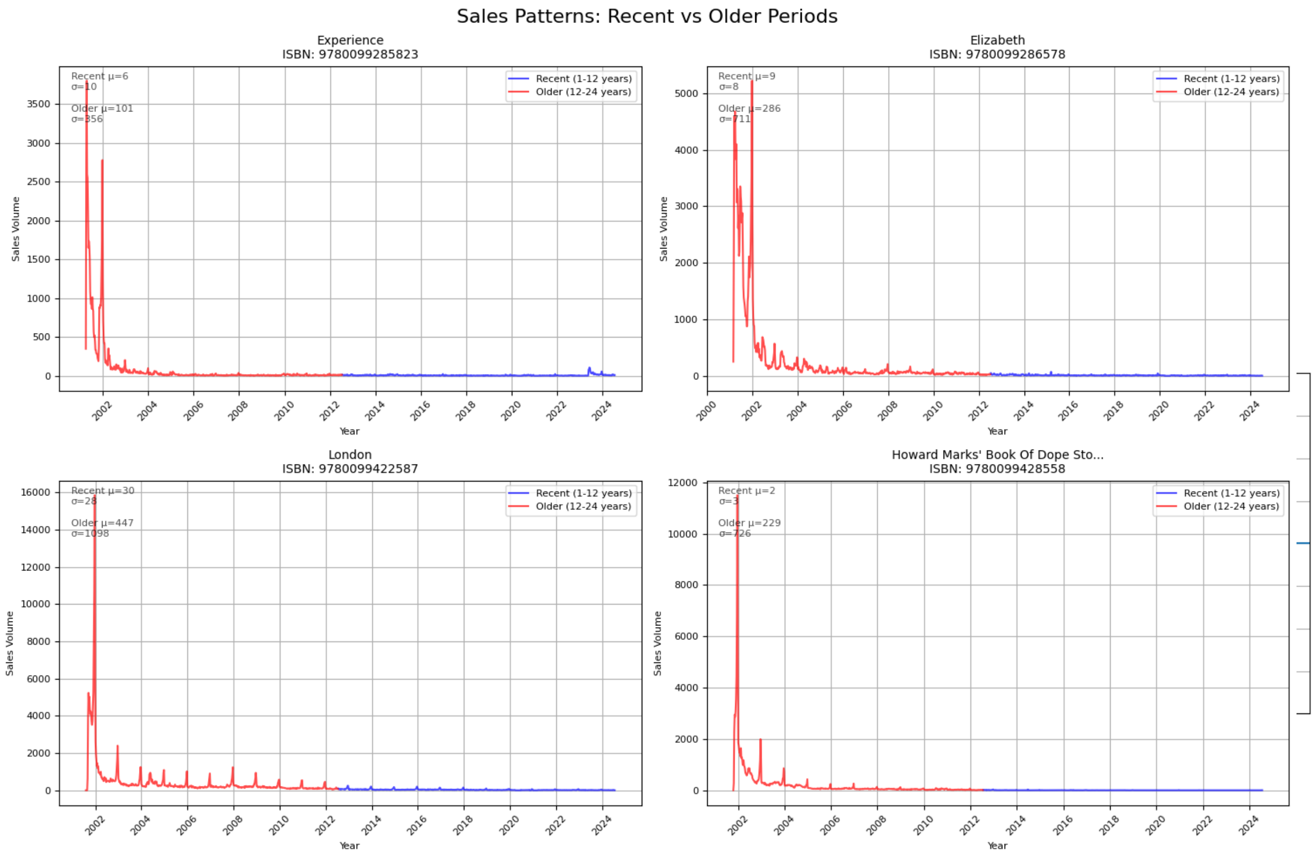
A.5: Sales Patterns



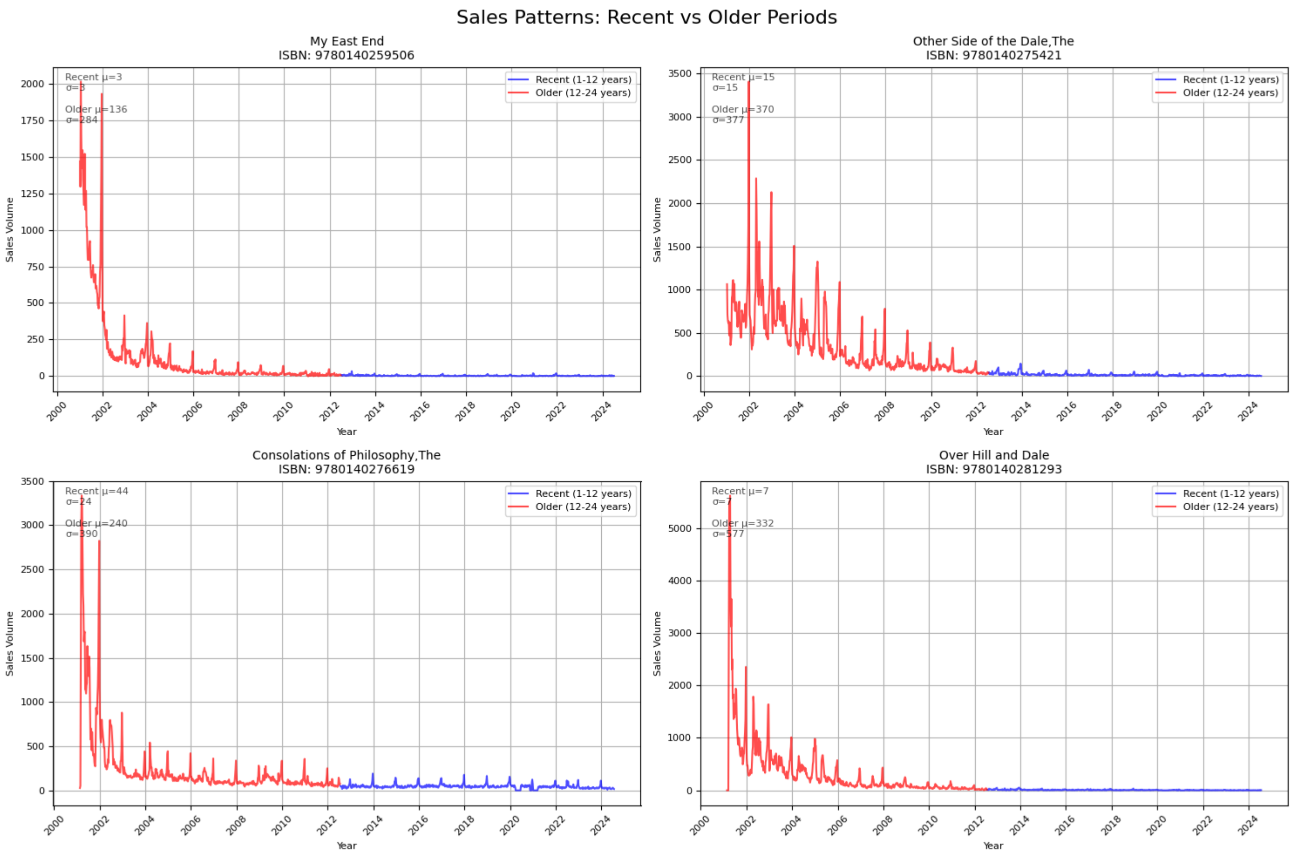
A.6: Sales Patterns



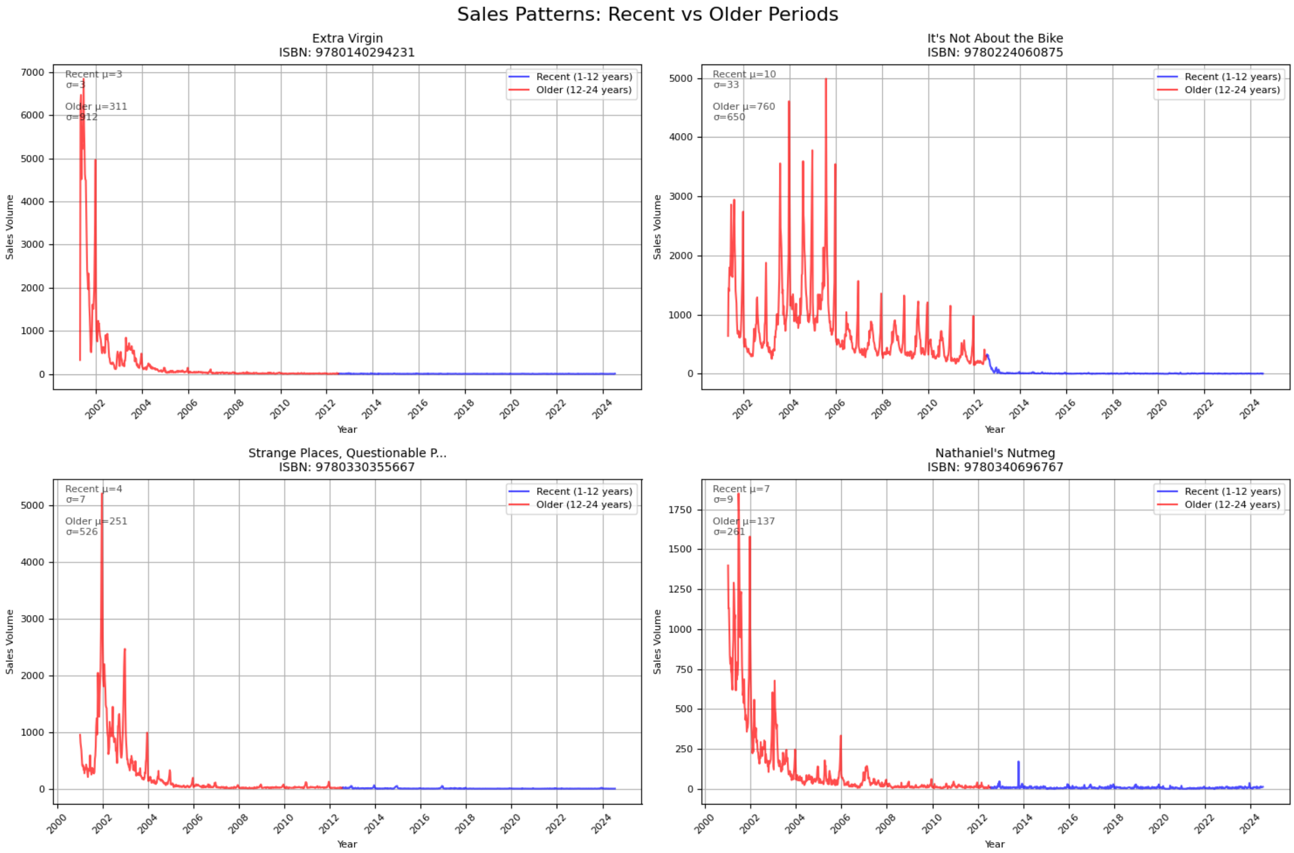
A.7: Sales Patterns



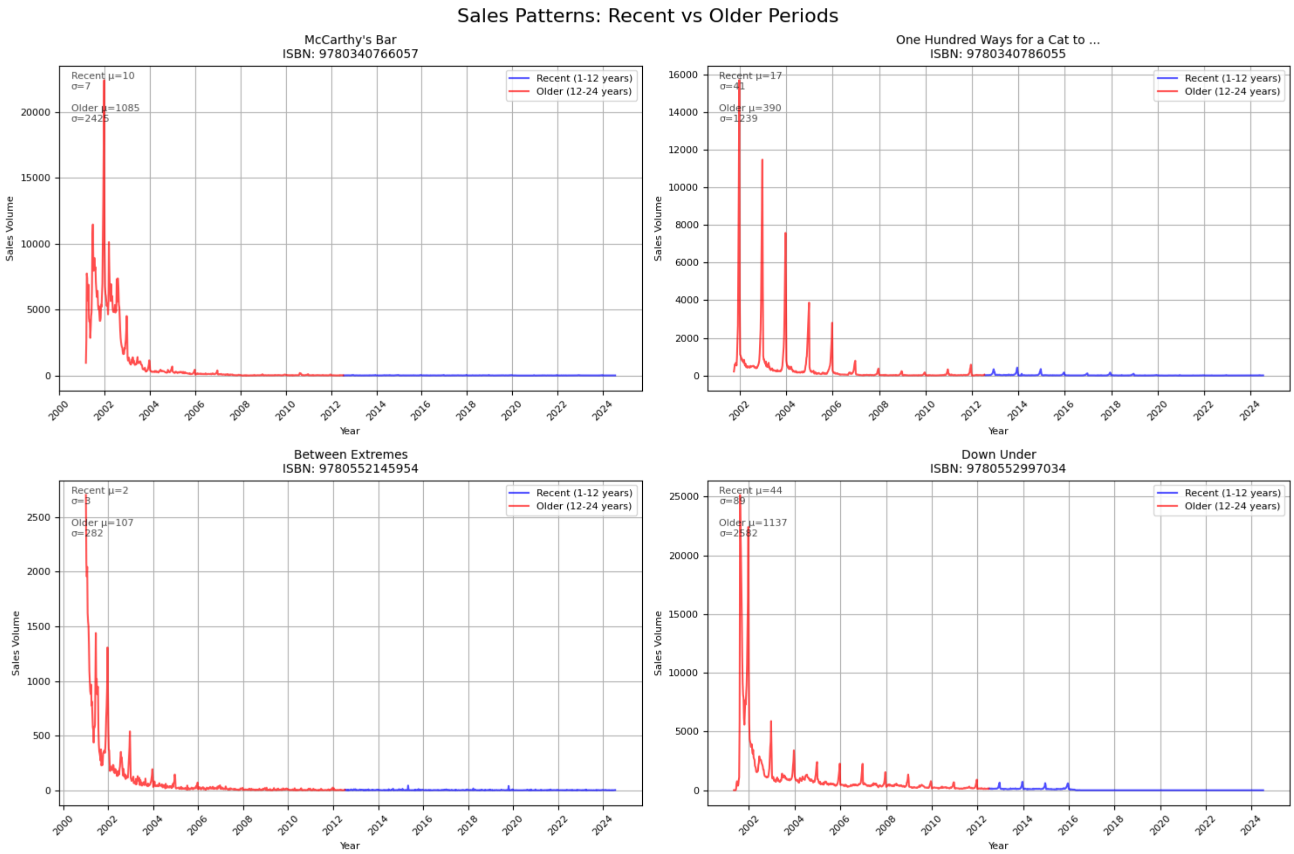
A.8: Sales Patterns



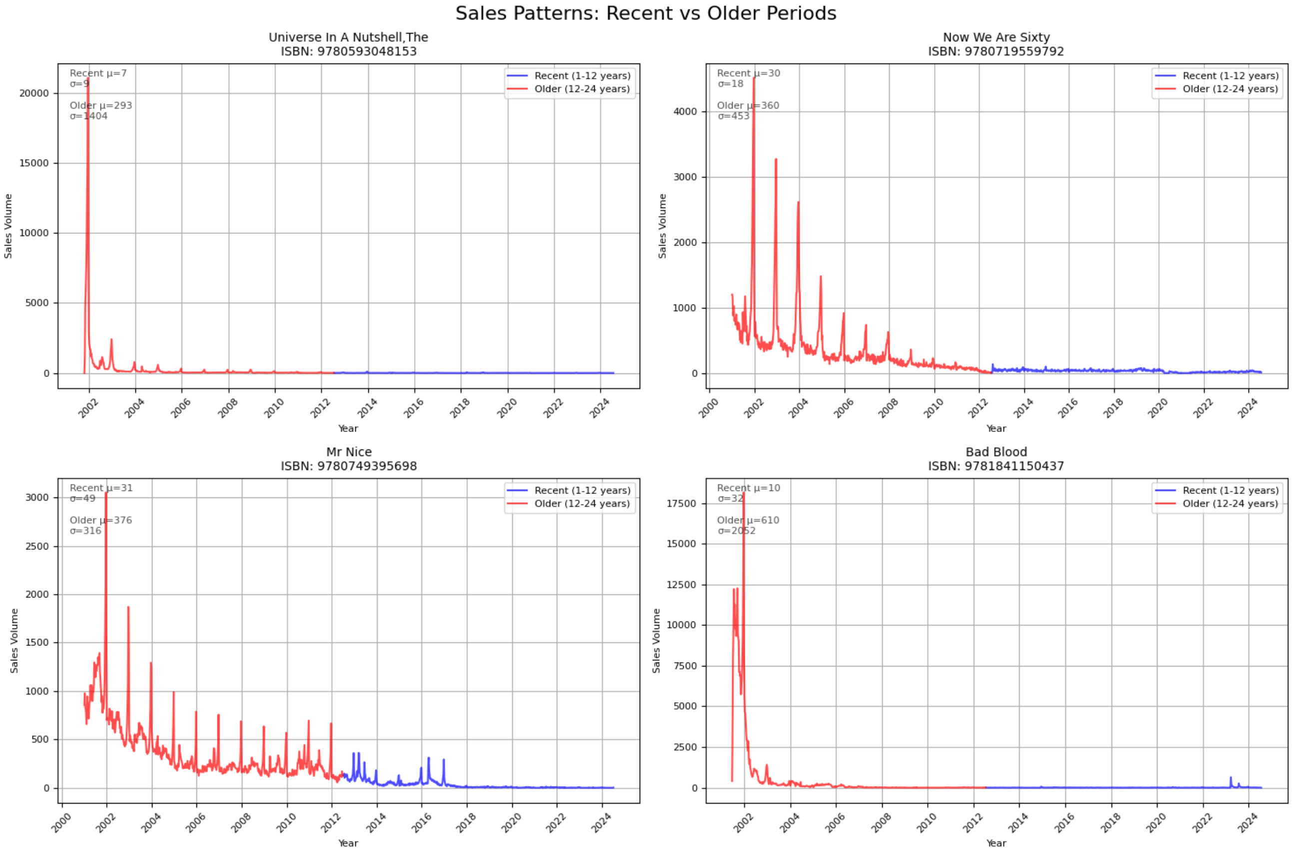
A.9: Sales Patterns



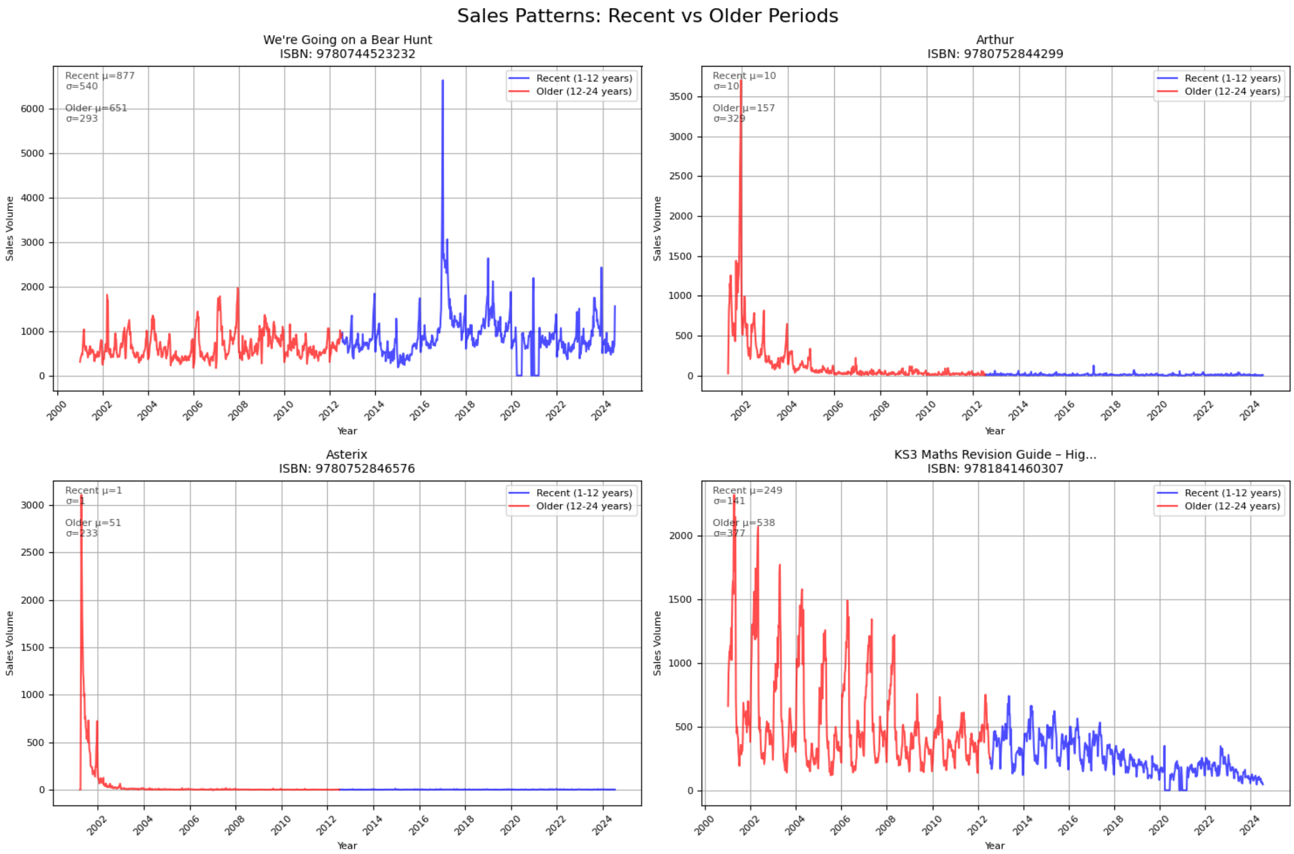
A.10: Sales Patterns



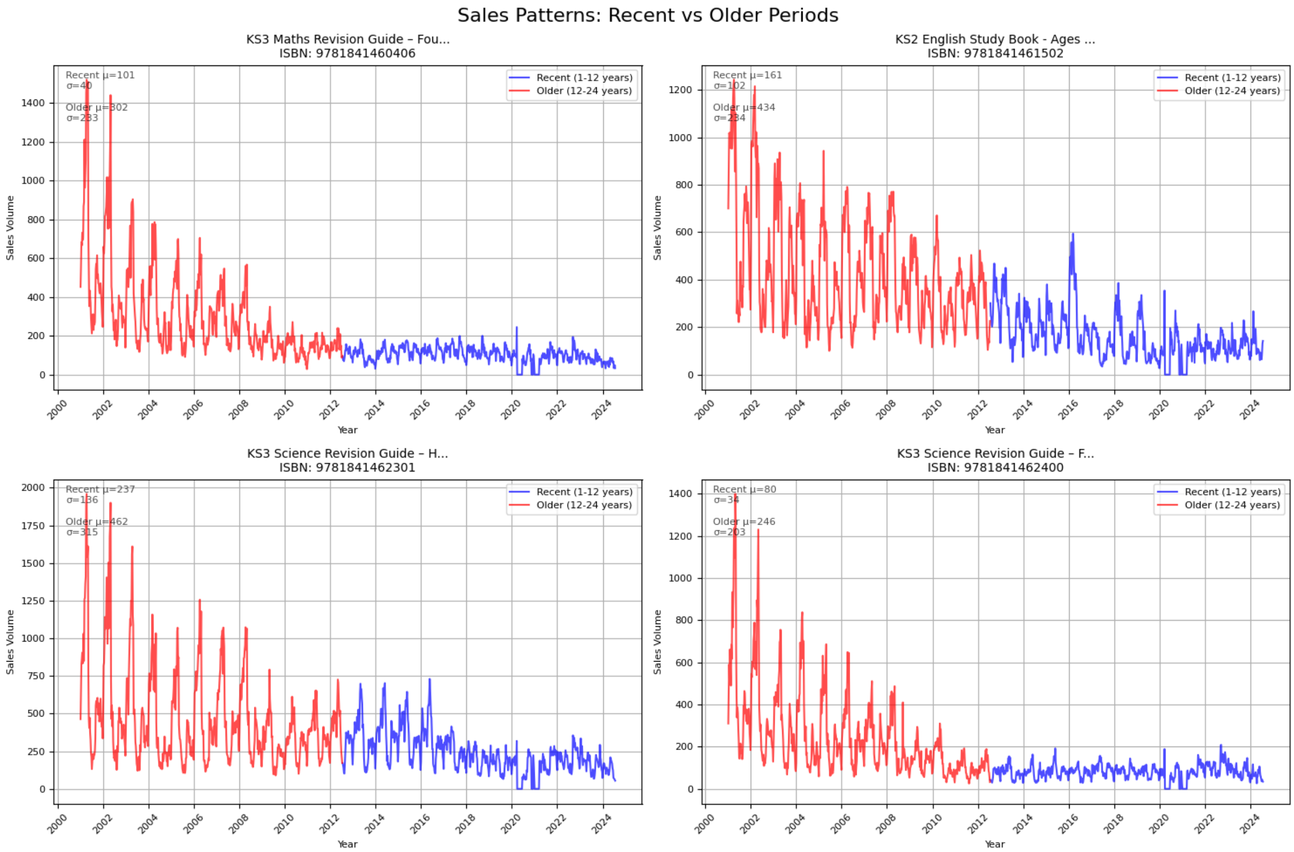
A.11: Sales Patterns



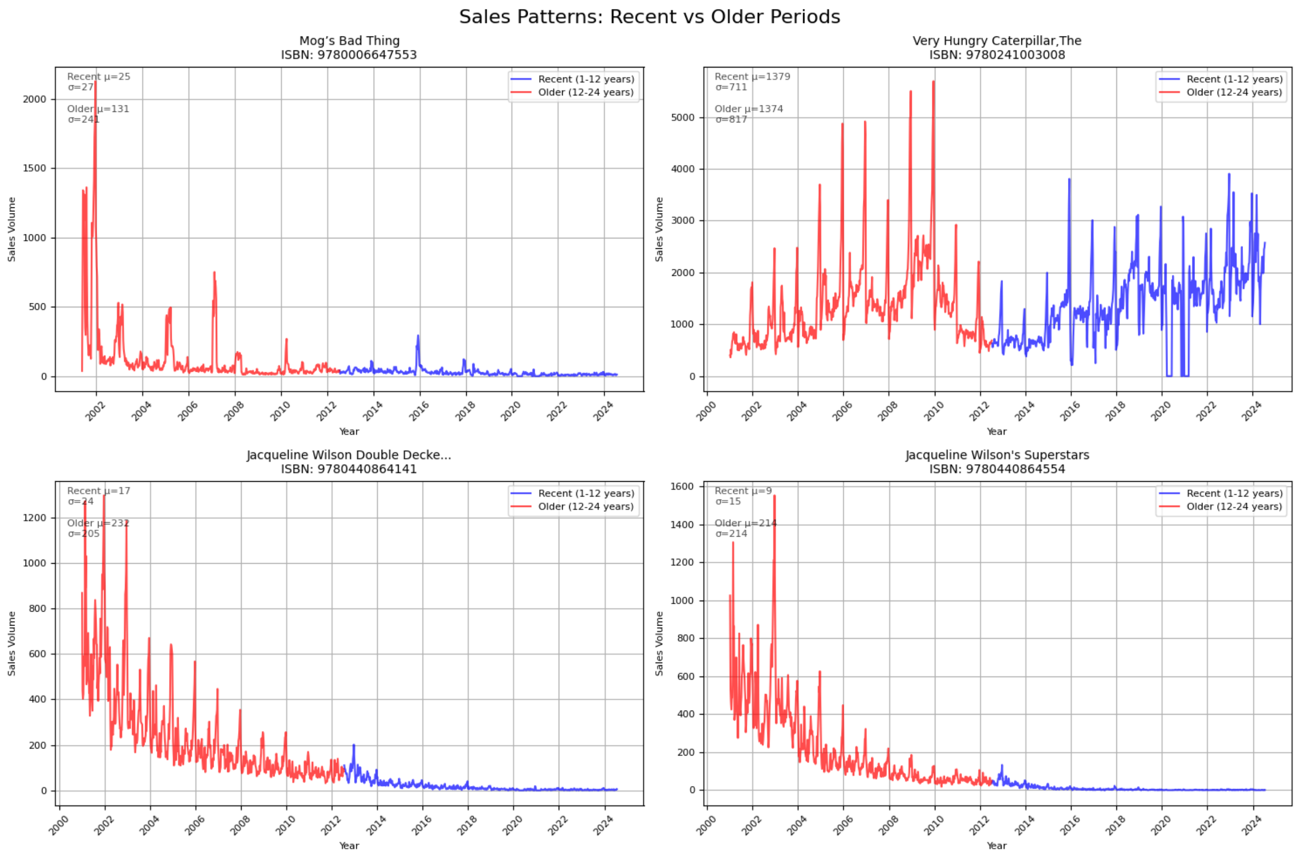
A.12: Sales Patterns



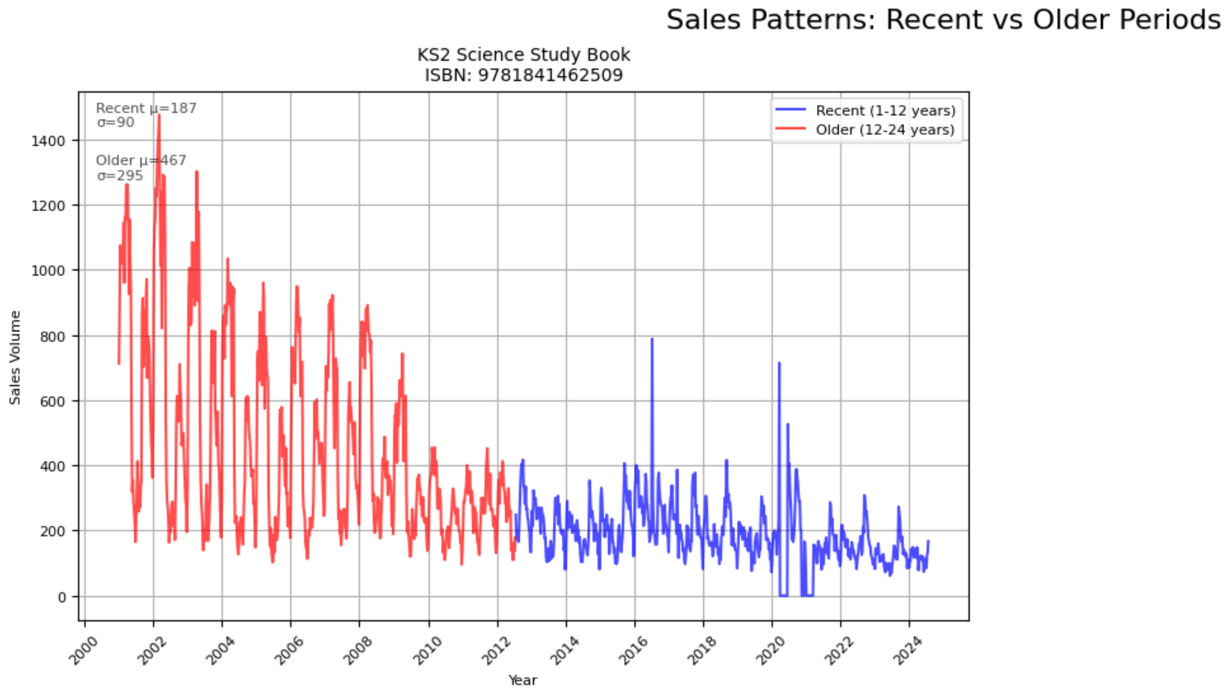
A.13: Sales Patterns



A.14: Sales Patterns

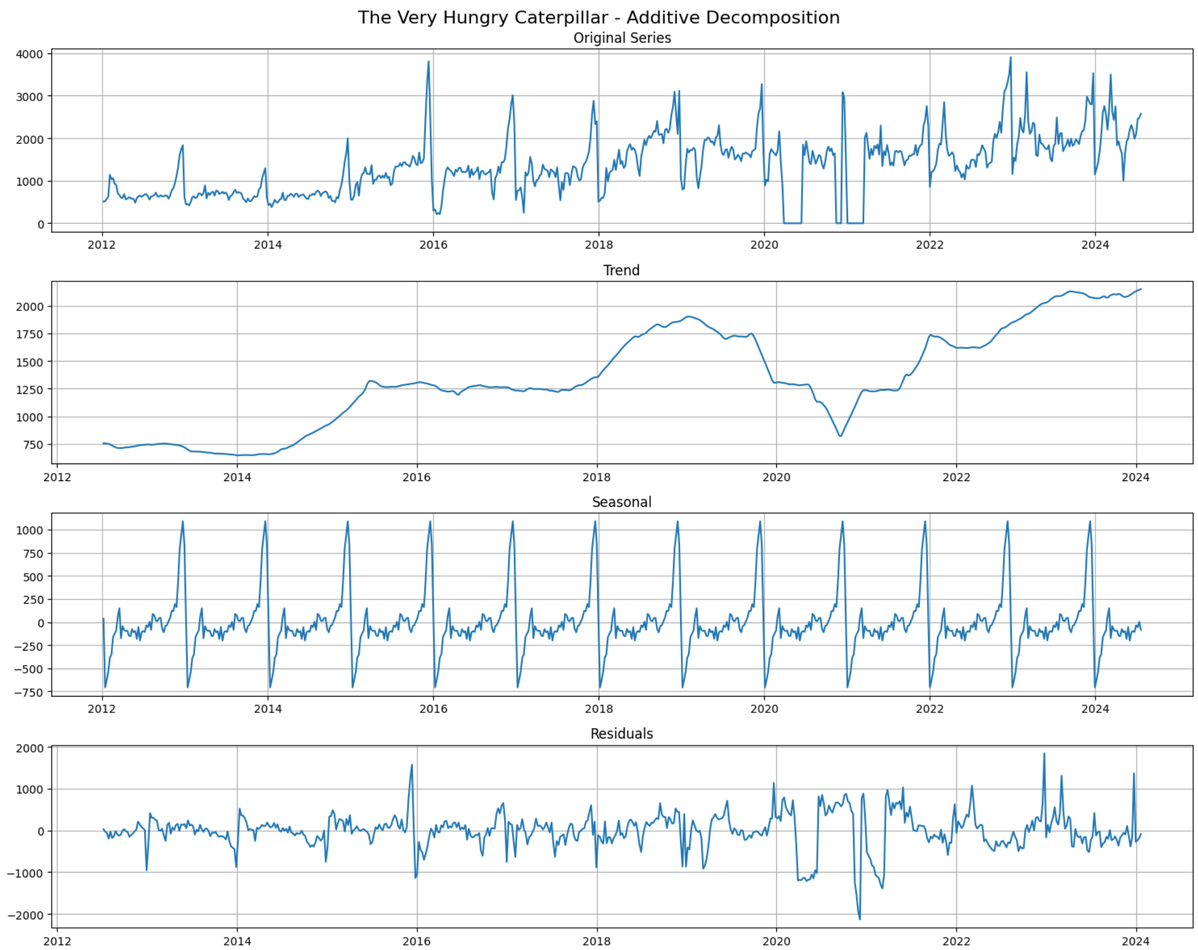


A.15: Sales Patterns

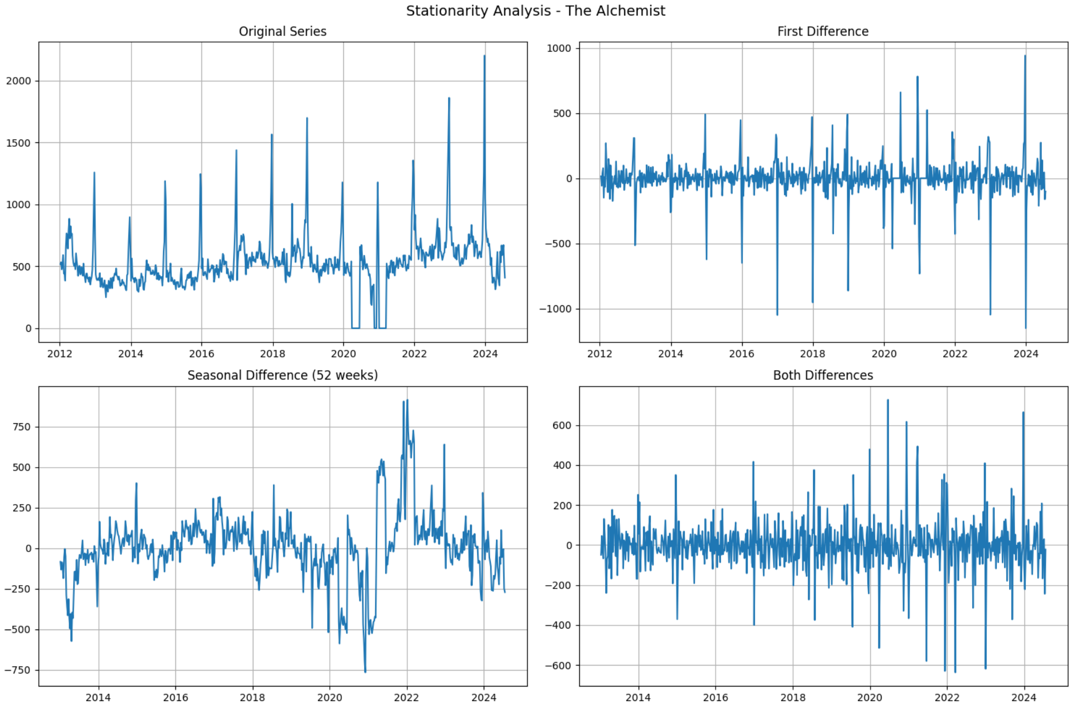


B: TIME SERIES DECOMPOSITION

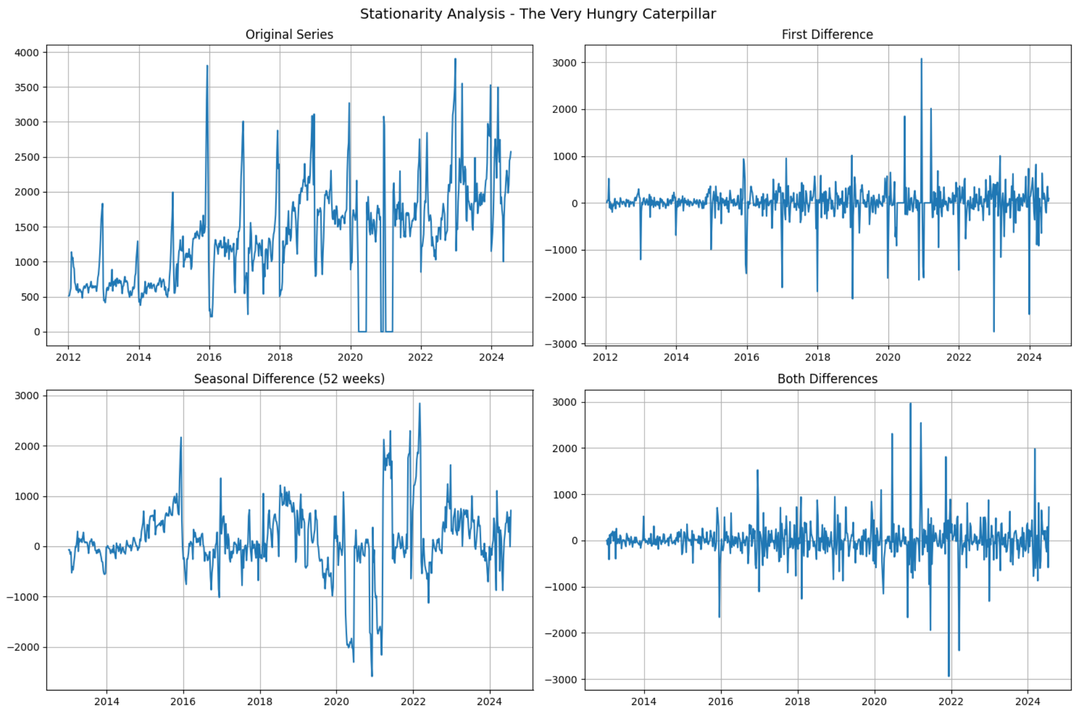
B.1: The Very Hungry Caterpillar Time series Decomposition Analysis



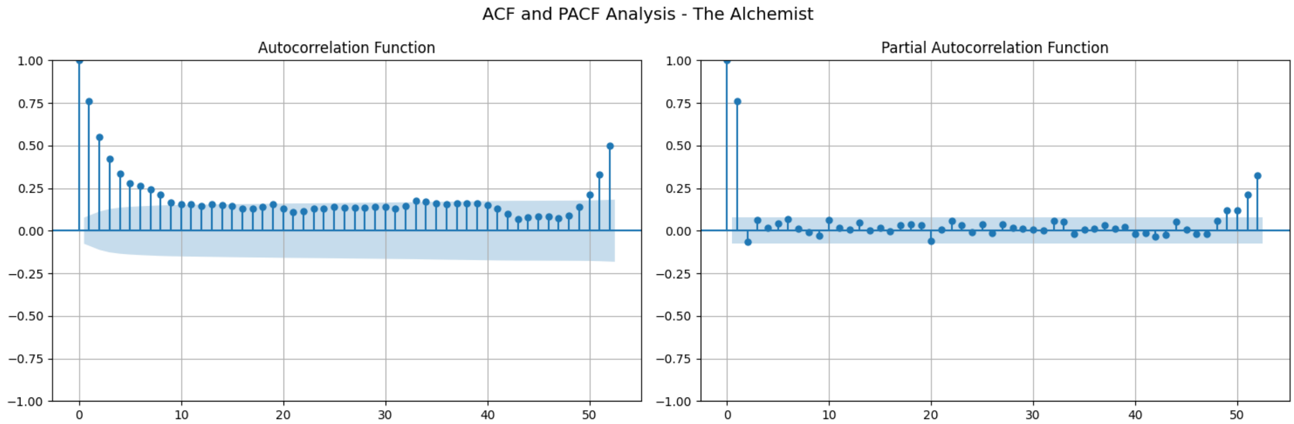
B.2: The Alchemist Stationarity



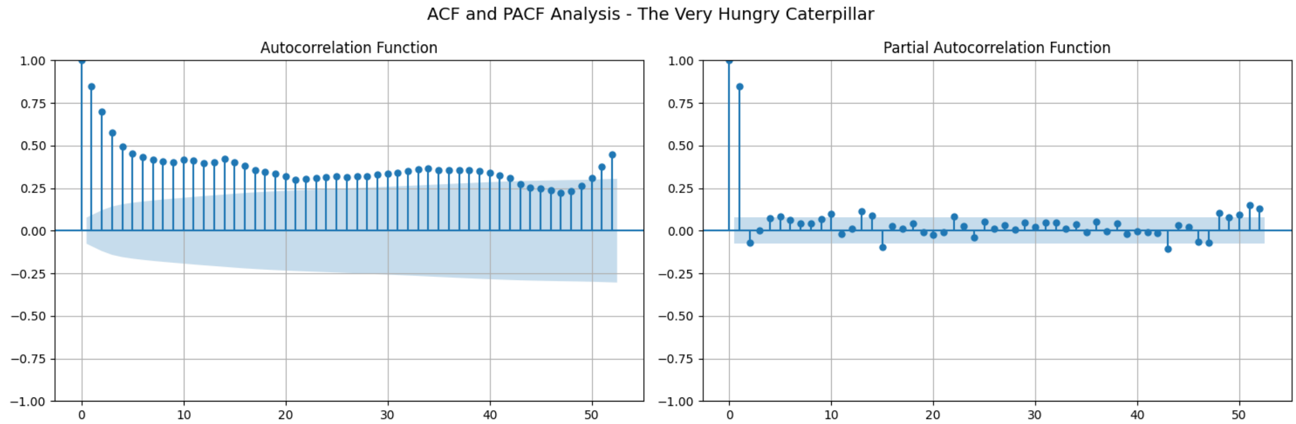
B.3: The Very Hungry Caterpillar Stationarity



B.4: The Alchemist ACF & PACF

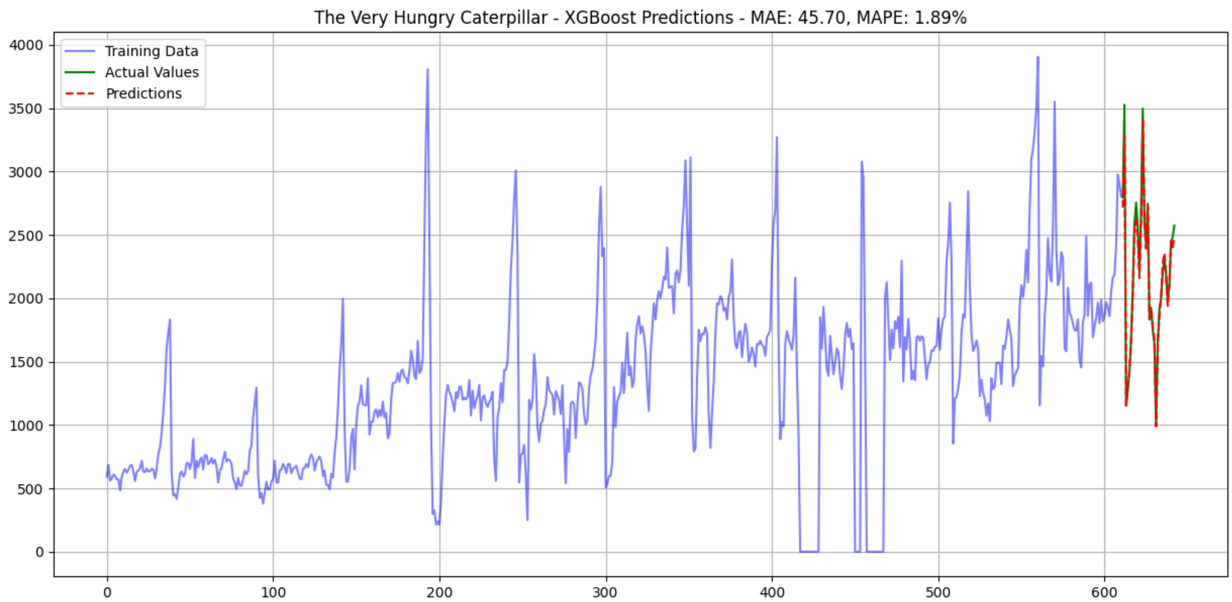


B.5: The Very Hungry Caterpillar ACF & PACF

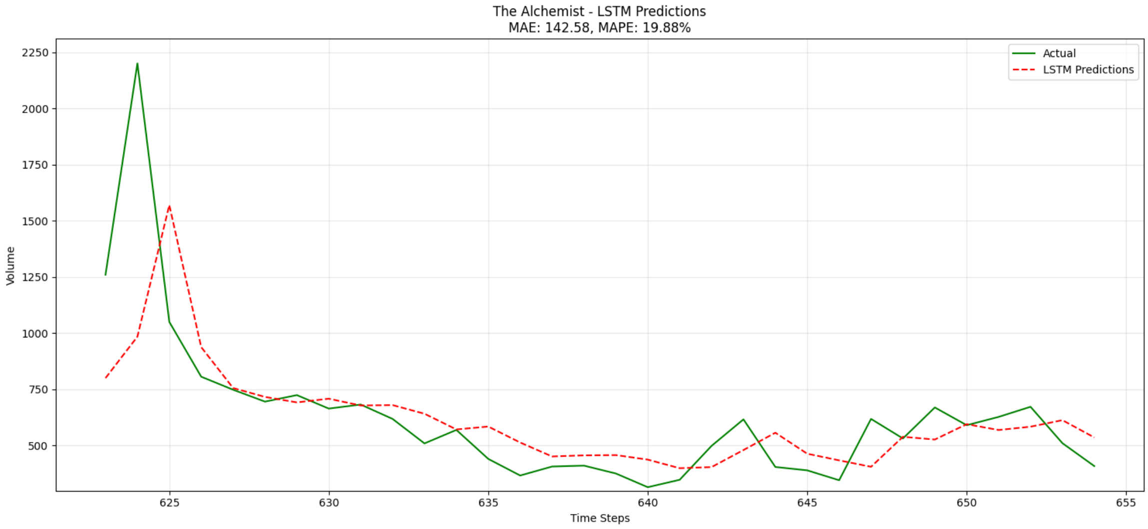


C: MACHINE LEARNING AND DEEP LEARNING

C1: The Very Hungry Caterpillar XGBoost Forecast



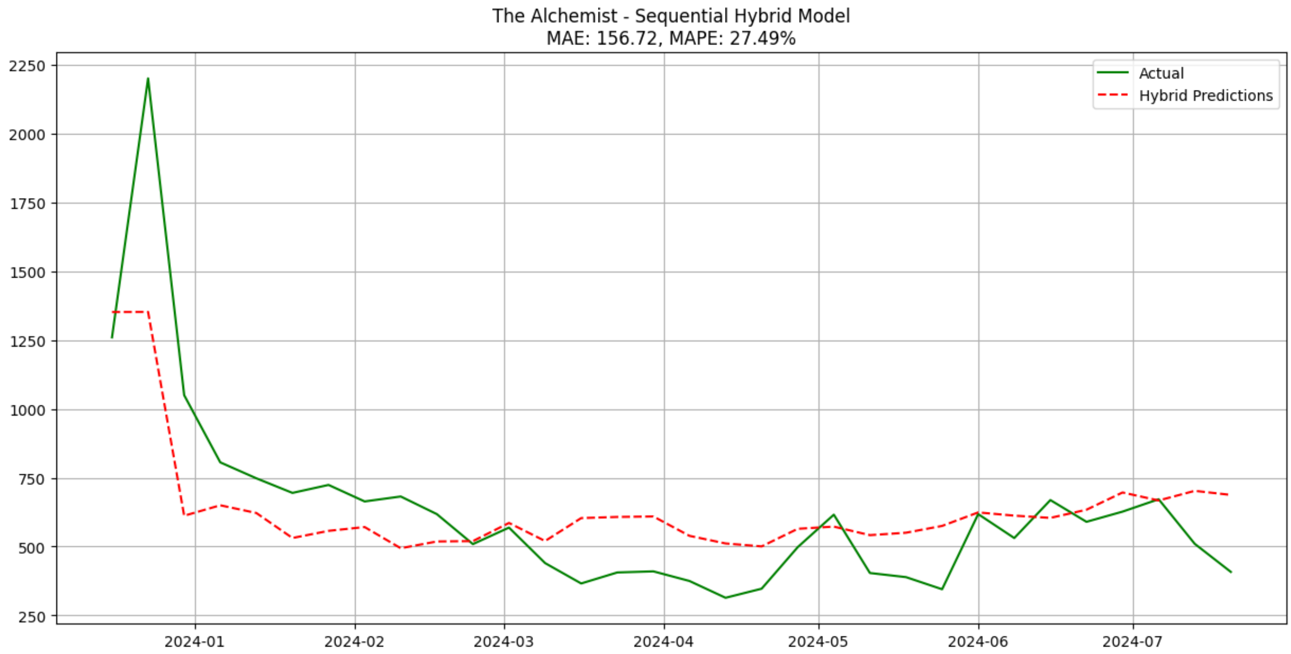
C2: The Alchemist - LSTM Forecast



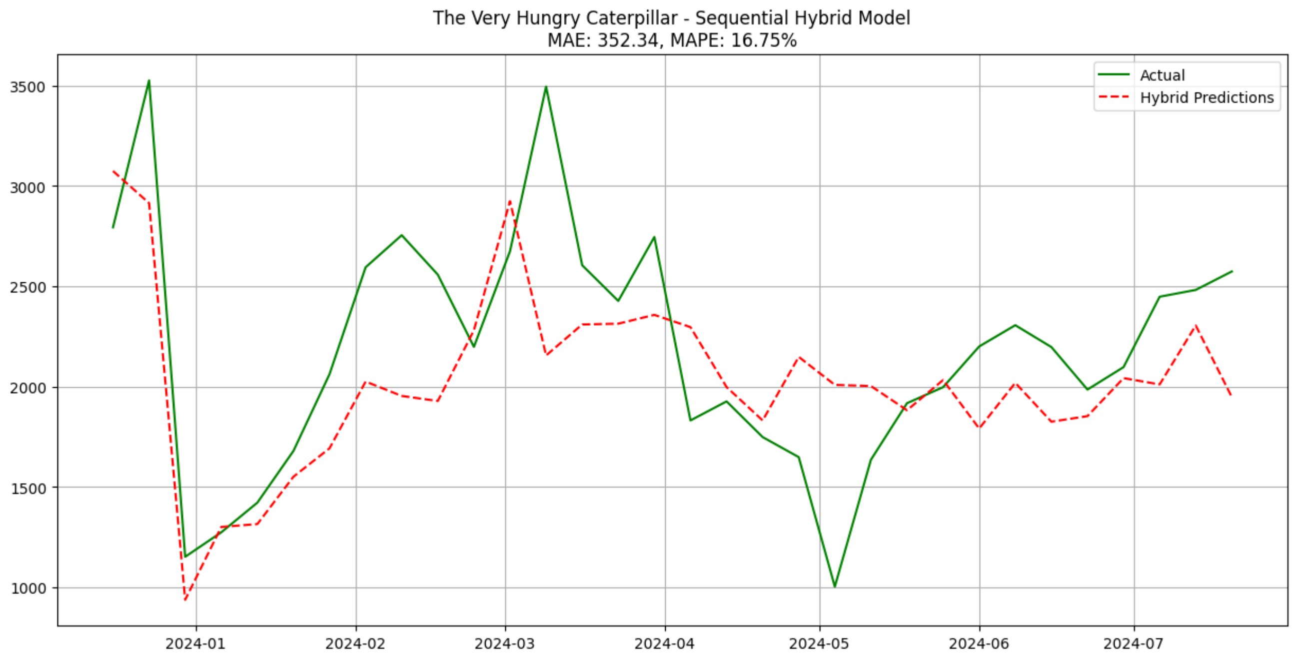
C3: The Very Hungry Caterpillar- LSTM Forecast

D: HYBRID MODELS

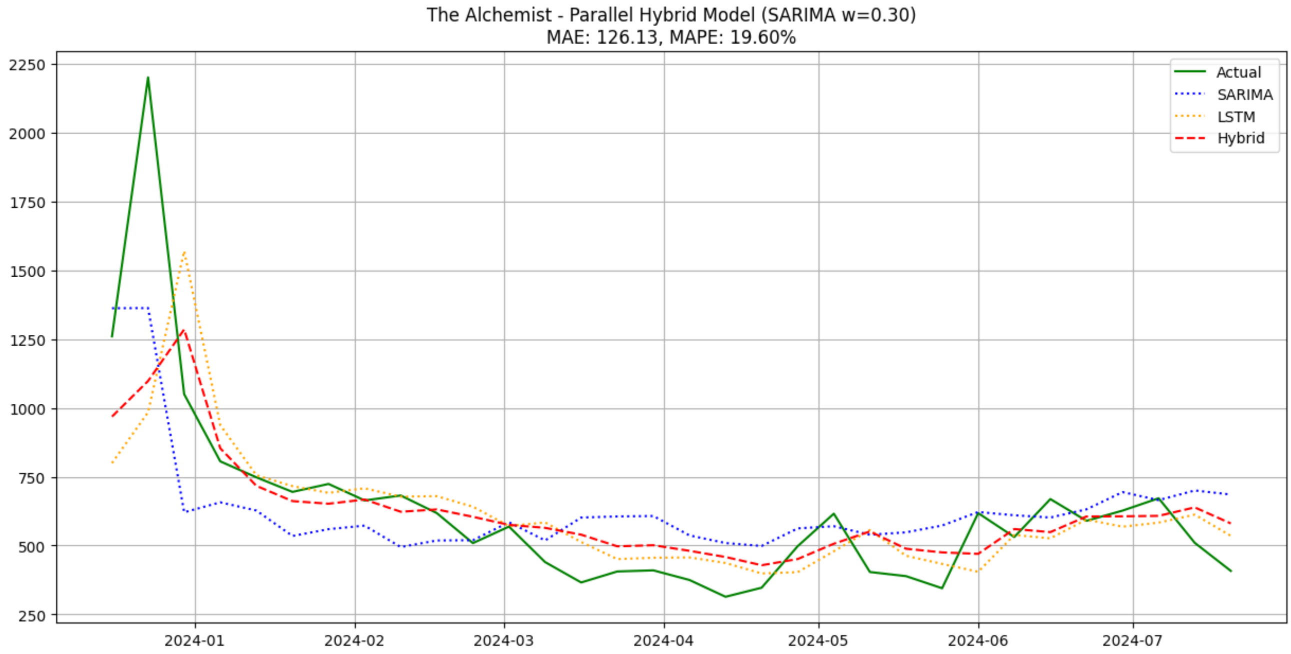
D1: The Alchemist - Sequential Hybrid



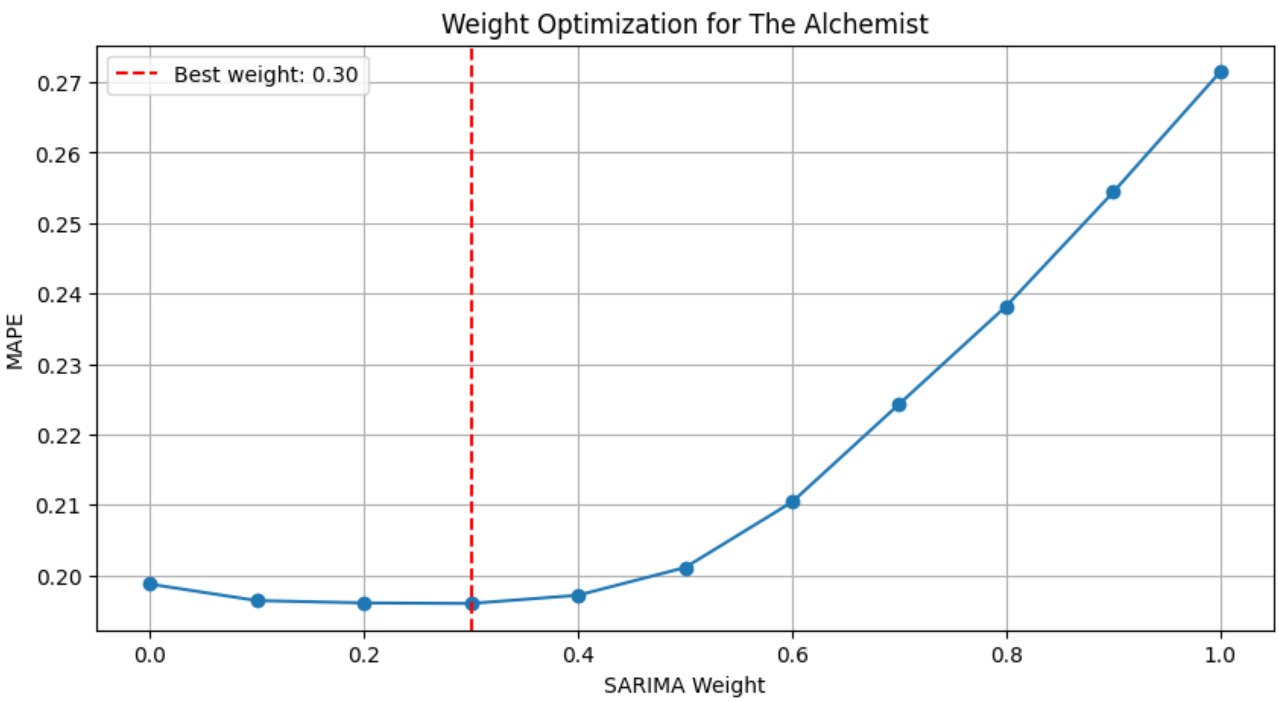
D2: The Very Hungry Caterpillar- Sequential Hybrid



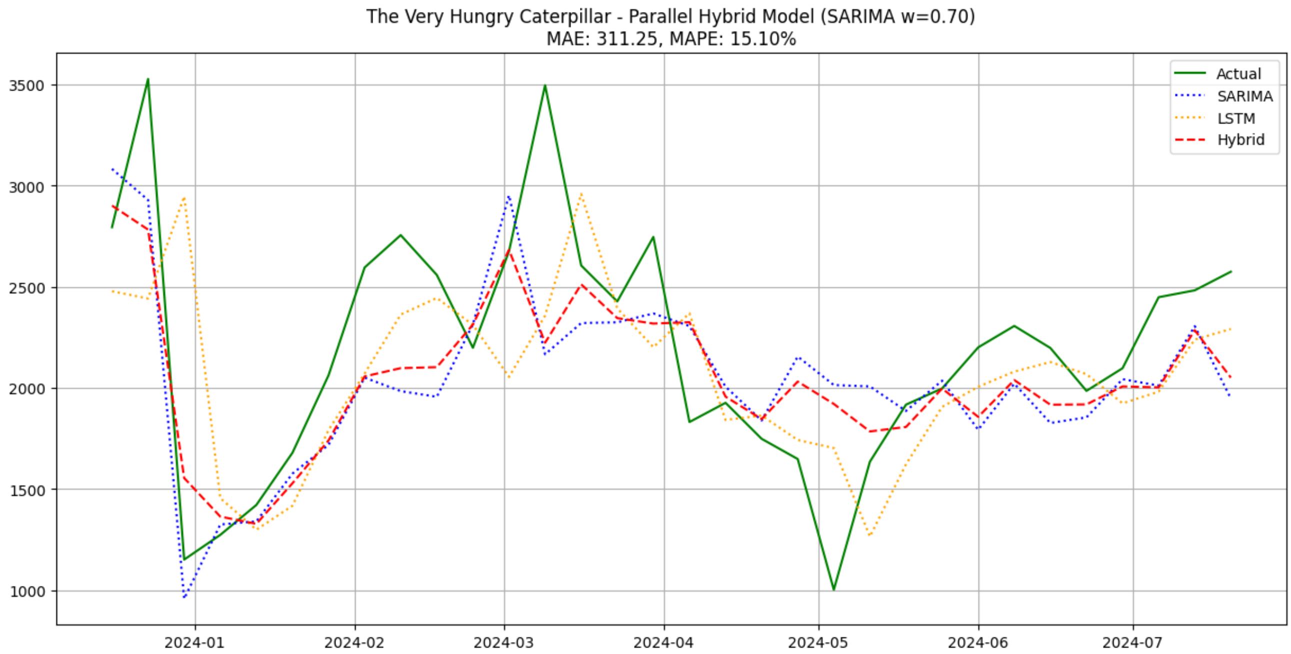
D3: The Alchemist -Parallel Hybrid



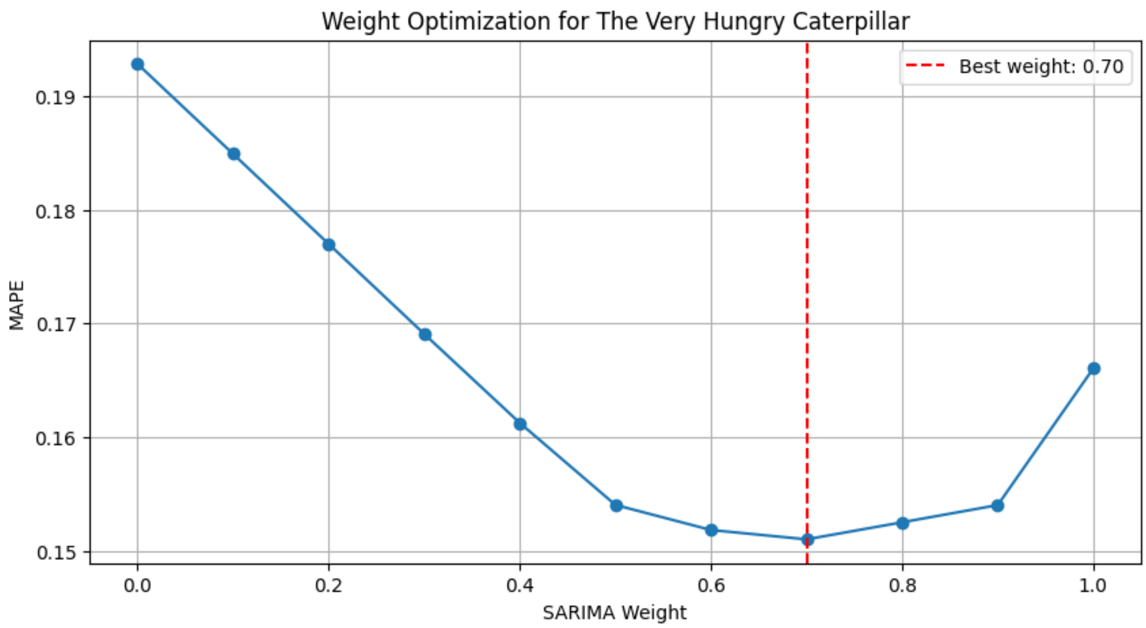
D4: The Alchemist - Parallel Hybrid - Weight Optimisation



D5: The Very Hungry Caterpillar- Parallel Hybrid

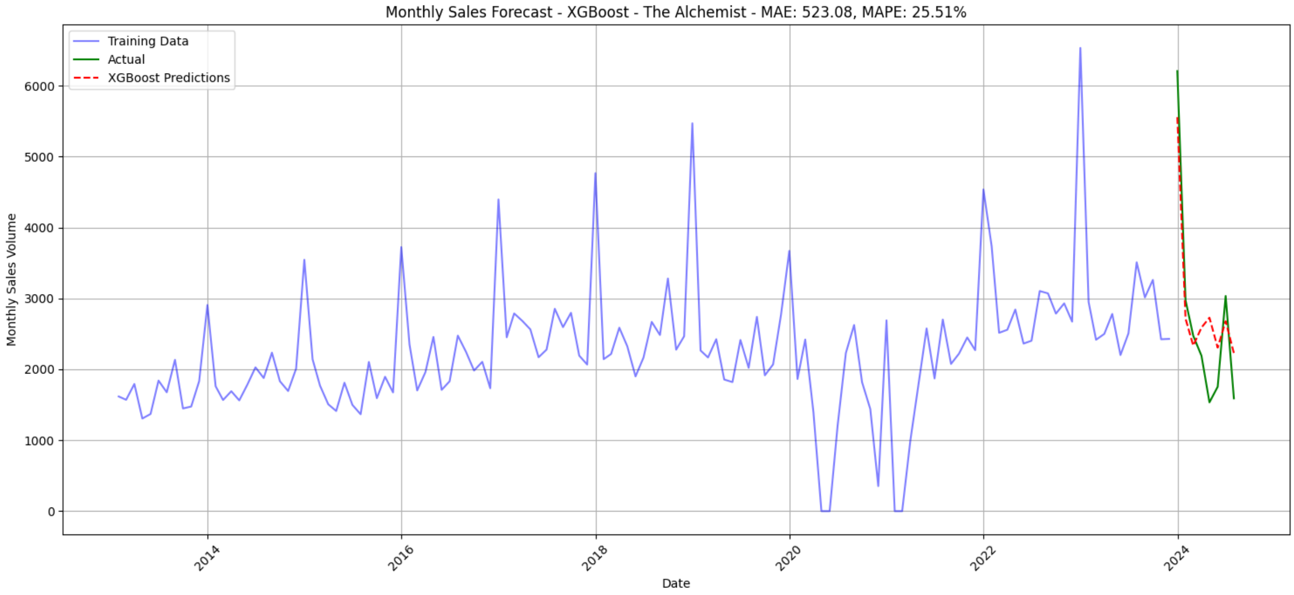


D6: The Very Hungry Caterpillar- Parallel Hybrid - Weight Optimisation

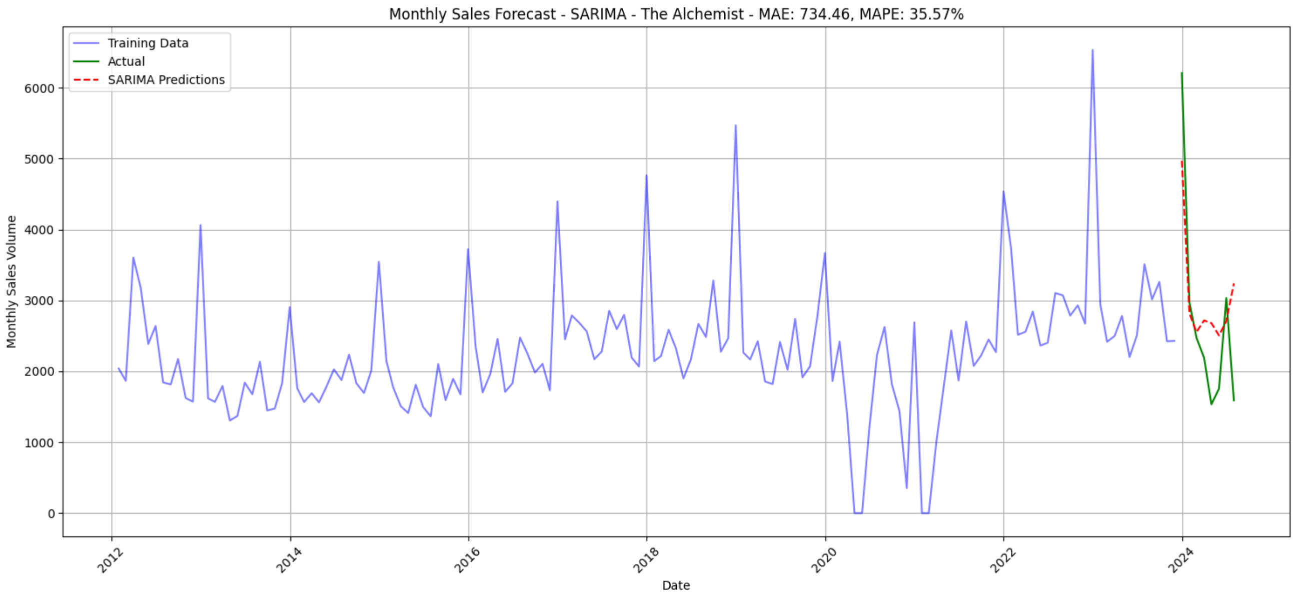


E: MONTHLY FORECASTING

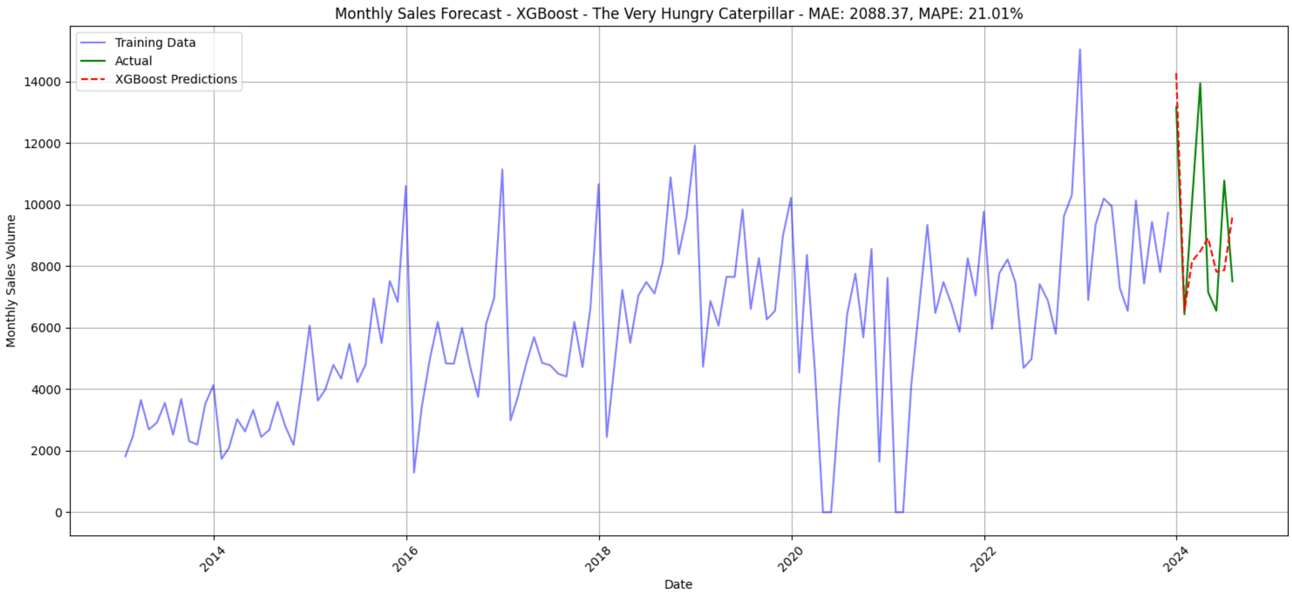
E1: The Alchemist - XGBoost Monthly Forecast



E2: The Alchemist - SARIMA Monthly Forecast



E3: The Very Hungry Caterpillar - XGBoost Monthly Forecast



E4: The Very Hungry Caterpillar - SARIMA Monthly Forecast